

Factoring an Integer with Three Oscillators and a Qubit

A Detailed Mathematical and Energetic Analysis

Report

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Abstract

We present a detailed analysis of the quantum factoring algorithm of Brenner, Caha, Coiteux-Roy, and König, which factors an integer N using only three bosonic oscillators and a single qubit, independent of the size of N . The algorithm achieves the same polynomial-time complexity as Shor's algorithm by replacing modular arithmetic with real arithmetic on continuous-variable position states, and the discrete Fourier transform with a homodyne P -quadrature measurement. After developing the relevant continuous-variable formalism, we derive the action of every gate in the circuit, analyze the idealized algorithm in full, and prove the correctness of the physical (finitely-squeezed) version with its error bounds. We also study the energetic cost of the algorithm: with the parameters specified by the paper, the energy required scales as $2^{\Theta(n^2)}$ where $n = \lceil \log_2 N \rceil$, rendering the protocol astronomically expensive in practice. We give explicit numerical estimates for factoring 3-digit and 8-digit integers. Detailed proofs and technical lemmas are collected in the appendices.

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1. Introduction

Shor’s algorithm [2] factors an integer N in polynomial time on a quantum computer with $O(\log N)$ qubits. Despite three decades of progress, scaling Shor’s algorithm to cryptographically relevant integers remains out of reach: each logical qubit is constructed from many physical qubits, and the total physical qubit count for factoring a 2048-bit RSA modulus is estimated to be in the millions.

A natural question is whether one can trade some of these qubits for continuous-variable (CV) resources. Brenner, Caha, Coiteux-Roy, and König answer this question affirmatively in a striking way [1]: *three bosonic oscillators and a single qubit suffice to factor N in polynomial time, regardless of the size of N .*

The price of this dramatic reduction in mode count is paid in another currency: the oscillators must be prepared in highly squeezed Gottesman–Kitaev–Preskill (GKP) states, and the required squeezing is enormous: the squeezing parameter grows as $\Theta(n^2)$ where $n = \lceil \log_2 N \rceil$, so the oscillator energy scales as $2^{\Theta(n^2)}$ — equivalently $N^{\Theta(\log N)}$, super-polynomial in N . The protocol is thus a proof of principle, not a practical pathway to factoring.

1.1. Key conceptual differences from Shor’s algorithm

The 3-oscillator-1-qubit algorithm differs from Shor’s at the level of how each ingredient of the original is realized in the CV setting:

- (i) **Registers are continuous, not discrete.** Position eigenstates $|x\rangle$ for $x \in \mathbb{R}$ replace finite-dimensional qubit registers.
- (ii) **Arithmetic is real, not modular.** The natural gates on an oscillator (squeezing, displacements) compute over $\mathbb{R}$. Modular structure is enforced *geometrically* by preparing one register on a lattice with spacing N .
- (iii) **The Fourier transform is implemented physically.** A homodyne P -quadrature measurement is a Fourier transform followed by a position measurement [3]; no separate QFT subroutine is needed.
- (iv) **The role of the qubit is minimal.** The qubit is briefly rented inside each elementary gate to hold a single bit of intermediate data and is always returned to $|0\rangle$.

1.2. The Shor dictionary

It is useful to keep in mind throughout the rest of the document the following explicit dictionary between Shor’s algorithm and its 3-oscillator-1-qubit analogue. Both algorithms find the period r of $f(x) = a^x \bmod N$; the differences are entirely in *how* this is done.

Role	Shor	3-osc-1-qubit
Input register	n qubits in $\sum_x x\rangle$	Oscillator A in $ \text{GKP}\rangle$
Output register	n qubits in $ 0\rangle$	Oscillator B in $M_N \text{GKP}\rangle$
Modular exponentiation	$U_{Z_N, a} : x\rangle y\rangle \rightarrow x\rangle (y + a^x) \bmod N\rangle$	$U_{a, N, m} : x\rangle y\rangle \rightarrow x\rangle y + f_{a, N, m}(x)\rangle$
Modular structure	Built into discrete arithmetic	Geometric: B's lattice has spacing N
Bit-by-bit multiplication	Quantum repeated squaring	V_α chain inside $V_{a, N, m}$
Period revealed by	QFT on input register	P -measurement on A
Output distribution	Peaks at $w = n/r, n \in \mathbb{Z}_q$	Delta-comb at $w = n/r, n \in \mathbb{Z}$
Period extraction	Continued fractions on w	Continued fractions on w
Workspace	Many ancilla qubits	Oscillator C + qubit Q (reset each V_α)

The two algorithms produce the *same* measurement distribution (up to discretization in Shor and approximation error in the physical CV version); the post-processing is identical. The difference is that Shor uses $O(n)$ qubits in two registers, whereas the CV version uses 4 modes (3 oscillators + 1 qubit) at the cost of squeezing whose energy scales as $2^{\Theta(n^2)}$.

1.3. Structure of the document

We begin in Section 2 with the continuous-variable formalism: oscillators, the squeezing and displacement operators, and the GKP state. Section 3 derives the action of every gate in the algorithm, in increasing order of complexity: primitive gates ($M_\alpha, e^{-iRP}, U^{\text{LSB}}$), then composite gates ($V_\alpha, V_{a, N, m}, U_{a, N, m}$). Section 4 analyzes the idealized algorithm assuming non-normalizable comb states. Section 5 introduces approximate (finitely squeezed) GKP states and proves correctness with explicit error bounds. Section 6 computes the energy budget of the algorithm and gives numerical estimates for small instances. The appendices contain detailed proofs of all technical lemmas.

2. Preliminaries

2.1. Hilbert space and canonical operators

Each oscillator lives in $L^2(\mathbb{R})$, the space of square-integrable functions of one real variable. The canonical position and momentum operators Q and P act in the position representation as

$$(Q\Psi)(x) = x\Psi(x), \quad (P\Psi)(x) = -i\frac{d\Psi}{dx}(x),$$

and satisfy the canonical commutation relation $[Q, P] = iI$.

Position eigenstates $|x\rangle$ are improper (non-normalizable) states satisfying $Q|x\rangle = x|x\rangle$ and $\langle x|x'\rangle = \delta(x - x')$. They are useful as a calculational basis but never appear as physical states by themselves. Their wavefunctions are formally $\langle y|x\rangle = \delta(y - x)$.

Momentum eigenstates $|\tilde{w}\rangle$ have wavefunctions $\langle x|\tilde{w}\rangle = e^{2\pi i w x}$, in the *frequency convention*, so that the Fourier transform $\text{FT}_{\mathbb{R}}$ on $L^2(\mathbb{R})$ takes the standard form $\hat{f}(w) = \int f(x) e^{-2\pi i w x} dx$. With $P = -i d/dx$ these satisfy $P|\tilde{w}\rangle = 2\pi w|\tilde{w}\rangle$; equivalently, the homodyne measurement we use reports the frequency $w = P/2\pi$. We work throughout in this frequency normalization, in which a position comb of spacing r has Fourier transform supported on $\frac{1}{r}\mathbb{Z}$. The qubit lives in $\mathbb{C}^2$ with basis $\{|0\rangle, |1\rangle\}$.

One-/two-mode Gaussian operation	Circuit representation	Definition
(Q -)displacement by ± 1	$ x\rangle \xrightarrow{e^{\mp i P}} x \mp 1\rangle$	$e^{\mp i P}$
Bosonic sum	$\begin{array}{c} x\rangle \\ y\rangle \end{array} \xrightarrow{\oplus} x+y\rangle$	$e^{-i Q_1 P_2}$
Single-mode squeezing	$ x\rangle \xrightarrow{S(z)} e^{z/2} e^{-z} \cdot x\rangle$	$e^{iz(QP+PQ)/2}$
(P -)quadrature measurement	$ \psi\rangle \xrightarrow{P} p$	$p \sim \hat{\psi}(p) ^2 dp$
Qubit-controlled Gaussian unitary	Circuit representation	Definition
(Q -)displacement by 1	$\begin{array}{c} b\rangle \\ x\rangle \end{array} \xrightarrow{\oplus} x+b\rangle$	$ 0\rangle\langle 0 \otimes I + 1\rangle\langle 1 \otimes e^{-i P}$
(Q -)displacement by -1	$\begin{array}{c} b\rangle \\ x\rangle \end{array} \xrightarrow{\ominus} x-b\rangle$	$ 0\rangle\langle 0 \otimes I + 1\rangle\langle 1 \otimes e^{i P}$
(P -)displacement by $\pm\pi$	$\begin{array}{c} b\rangle \\ x\rangle \end{array} \xrightarrow{e^{\pm i \pi Q}} e^{\pm i \pi x b} x\rangle$	$ 0\rangle\langle 0 \otimes I + 1\rangle\langle 1 \otimes e^{\pm i \pi Q}$
Phase-space rotation by $\pm\pi/2$	$\begin{array}{c} b\rangle \\ \psi\rangle \end{array} \xrightarrow{e^{\pm i \pi \hat{N}/2}} e^{\pm i \pi b \hat{N}/2} \psi\rangle$	$ 0\rangle\langle 0 \otimes I + 1\rangle\langle 1 \otimes e^{\pm i \pi \hat{N}/2}$

Figure 1: The elementary single- and two-mode Gaussian operations (top) and their qubit-controlled versions (bottom), with circuit representations and definitions. These are the hardware-level primitives from which all gates in Section 3 are built. Reproduced from [1].

The full Hilbert space of the algorithm is

$$\mathcal{H} = L^2(\mathbb{R})_A \otimes L^2(\mathbb{R})_B \otimes L^2(\mathbb{R})_C \otimes \mathbb{C}_Q^2.$$

2.2. The vacuum and squeezed states

In quantum optics, a single mode of the electromagnetic field is described by a harmonic oscillator. The natural transformations one can apply are Gaussian operations: displacements (shifting the state in position or momentum), phase-space rotations, and *squeezings* (scaling one quadrature while inversely scaling the other). The squeezing operator is the central primitive: applied to position eigenstates, it implements multiplication by a real number, which is the fundamental arithmetic operation used to build the algorithm. The elementary Gaussian operations available on these modes—together with their qubit-controlled versions—are collected in Figure 1.

Definition 2.1: Vacuum state

The vacuum $|\text{vac}\rangle$ is the unique normalized state annihilated by $a = (Q + iP)/\sqrt{2}$. Its wavefunction is

$$\Psi_{\text{vac}}(x) = \pi^{-1/4} e^{-x^2/2}.$$

(Derivation in Appendix A.1.) This is a Gaussian of width 1 centered at 0, the minimum-uncertainty state.

Definition 2.2: Squeezing operator

For $z \in \mathbb{R}$, define

$$S(z) := e^{iz(QP+PQ)/2}.$$

The squeezing operator is unitary and squeezes one quadrature while stretching the other. Its action on canonical operators is summarized in the following lemma, which is central to everything else.

Lemma 2.3: Heisenberg action of $S(z)$

For any $z \in \mathbb{R}$:

$$S(z) Q S(z)^\dagger = e^z Q, \quad S(z) P S(z)^\dagger = e^{-z} P.$$

Proof. We use the Baker–Campbell–Hausdorff formula $e^A B e^{-A} = \sum_{k \geq 0} \frac{1}{k!} \text{ad}_A^k(B)$, where $\text{ad}_A(B) := [A, B]$. Set $A = iz(QP + PQ)/2$ and $B = Q$.

We compute $[A, Q]$:

$$[QP, Q] = Q[P, Q] + [Q, Q]P = -iQ, \quad [PQ, Q] = P[Q, Q] + [P, Q]Q = -iQ,$$

so $[A, Q] = \frac{iz}{2}(-2iQ) = zQ$. Iterating, $\text{ad}_A^k(Q) = z^k Q$, so

$$S(z) Q S(z)^\dagger = \sum_{k \geq 0} \frac{z^k}{k!} Q = e^z Q.$$

The proof for P is analogous, using $[QP, P] = iP$ and $[PQ, P] = iP$, giving $[A, P] = -zP$. $\square$

Corollary 2.4: Action of $S(z)$ on position eigenstates

$S(z) |x'\rangle = |e^{-z}x'\rangle$ for any $x' \in \mathbb{R}$.

Proof. Compute $Q(S(z) |x'\rangle)$ by inserting $I = S(z)S(z)^\dagger$:

$$Q S(z) |x'\rangle = S(z) (S(z)^\dagger Q S(z)) |x'\rangle = S(z) (e^{-z} Q) |x'\rangle = e^{-z} x' \cdot S(z) |x'\rangle,$$

where we used Lemma 2.3 (with $z \rightarrow -z$) to get $S(z)^\dagger Q S(z) = e^{-z} Q$. Thus $S(z) |x'\rangle$ is a Q -eigenstate with eigenvalue $e^{-z}x'$, i.e. $|e^{-z}x'\rangle$. $\square$

2.3. The scalar multiplication gate M_α

Definition 2.5: Scalar multiplication gate

For $\alpha > 0$, define $M_\alpha := S(-\log \alpha)$.

Lemma 2.6

$M_\alpha |x\rangle = |\alpha x\rangle$ for any $x \in \mathbb{R}$. On wavefunctions, $(M_\alpha \Psi)(x) = \frac{1}{\sqrt{\alpha}} \Psi(x/\alpha)$.

Proof. By Corollary 2.4 with $z = -\log \alpha$: $M_\alpha |x\rangle = |e^{\log \alpha} x\rangle = |\alpha x\rangle$. The wavefunction form follows by computing $(M_\alpha \Psi)(x) = \langle x | M_\alpha \Psi \rangle$ and inserting a resolution of the identity (see Appendix A.2). $\square$

The exponents in $S(z)$ commute, so

$$S(z_1)S(z_2) = S(z_1 + z_2), \quad (1)$$

which allows large-strength squeezings to be decomposed into constant-strength ones:

$$M_\alpha = S(-\log \alpha) = S(-\log \alpha / \ell)^\ell.$$

Choosing $\ell = \lceil |\log \alpha| \rceil$ ensures each elementary squeezing has strength at most 1. This is the technologically motivated reason for the decomposition: large squeezings are physically difficult.

2.4. The displacement operator e^{-iRP}

Lemma 2.7: Heisenberg action of e^{-iRP}

For $R \in \mathbb{R}$:

$$e^{iRP} Q e^{-iRP} = Q + R \cdot I, \quad e^{iRP} P e^{-iRP} = P.$$

Proof. Set $A = iRP$. Then $[A, Q] = iR[P, Q] = iR(-iI) = RI$, and $[A, RI] = 0$, so the BCH series terminates: $e^A Q e^{-A} = Q + RI$. Similarly $[A, P] = 0$, giving $e^A P e^{-A} = P$. $\square$

Corollary 2.8

$e^{-iRP} |x\rangle = |x + R\rangle$ for $x, R \in \mathbb{R}$.

Proof. $Q e^{-iRP} |x\rangle = e^{-iRP} (e^{iRP} Q e^{-iRP}) |x\rangle = e^{-iRP} (Q + RI) |x\rangle = (x + R) e^{-iRP} |x\rangle$. $\square$

For unit displacement, the wavefunction action is $(e^{iP} \Psi)(x) = \Psi(x + 1)$, derived in Appendix A.3 via the Taylor series identity $e^{d/dx} \Psi(x) = \Psi(x + 1)$.

2.5. Approximate position eigenstates and GKP states

Position eigenstates $|x\rangle$ are not physical because they have infinite energy (an infinitely localized wavefunction has infinite momentum spread). For physical computation we need normalizable analogues. Similarly, the idealized uniform superposition $\sum_{x \in \mathbb{Z}} |x\rangle$ over integer positions —

which we shall see plays the role of $|+\rangle^{\otimes n}$ in Shor's algorithm — is not a normalizable state, so we use a finitely-squeezed approximation: the Gottesman-Kitaev-Preskill (GKP) state [3], a sum of Gaussian peaks at integer positions, weighted by a Gaussian envelope of finite width.

We will need normalizable analogues of $|x\rangle$ and $|\text{GKP}\rangle$.

Definition 2.9: Approximate position eigenstate

For $z \in \mathbb{R}$ and $\Delta > 0$:

$$|\chi_\Delta(z)\rangle := e^{-iPz} S(\log(1/\Delta)) |\text{vac}\rangle.$$

Equivalently, $|\chi_\Delta(z)\rangle = e^{-iPz} M_\Delta |\text{vac}\rangle$. The wavefunction is a Gaussian of width Δ centered at z :

$$\chi_\Delta(z)(x) = \frac{1}{(\pi\Delta^2)^{1/4}} e^{-(x-z)^2/(2\Delta^2)}.$$

As $\Delta \rightarrow 0$ this approaches the Dirac delta at z .

Definition 2.10: Approximate GKP state

For $\kappa, \Delta > 0$:

$$|\text{GKP}_{\kappa,\Delta}\rangle := C_{\kappa,\Delta} \sum_{z \in \mathbb{Z}} \eta_\kappa(z) |\chi_\Delta(z)\rangle,$$

where $\eta_\kappa(z) = \frac{\sqrt{\kappa}}{\pi^{1/4}} e^{-\kappa^2 z^2/2}$ is a Gaussian envelope of width $\sim 1/\kappa$ (normalized so that $\int |\eta_\kappa|^2 = 1$), and $C_{\kappa,\Delta}$ is a normalization constant.

In the limit $\kappa, \Delta \rightarrow 0$ the state $|\text{GKP}_{\kappa,\Delta}\rangle$ approaches the ideal Dirac comb $\sum_{z \in \mathbb{Z}} |z\rangle$; its comb-of-Gaussians structure is shown in Figure 2.

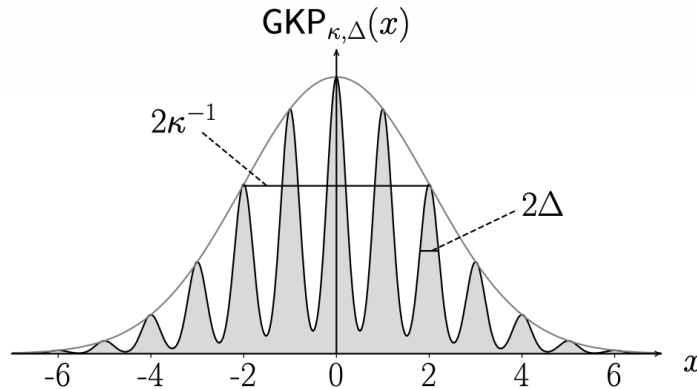


Figure 2: An approximate GKP state $|\text{GKP}_{\kappa,\Delta}\rangle$: a comb of Gaussian peaks of width $\sim \Delta$ under a Gaussian envelope of width $\sim 1/\kappa$. As $\kappa, \Delta \rightarrow 0$ it tends to the ideal Dirac comb $\sum_{z \in \mathbb{Z}} |z\rangle$. Reproduced from [1].

Definition 2.11: Truncated approximate GKP state

For $\varepsilon \in (0, 1/2)$, define the projector $\Pi_{\mathbb{Z}(\varepsilon)} = \int_{\mathbb{Z}(\varepsilon)} |x\rangle\langle x| dx$, where $\mathbb{Z}(\varepsilon) = \bigcup_{n \in \mathbb{Z}} [n - \varepsilon, n + \varepsilon]$, and set

$$|\text{GKP}_{\kappa,\Delta}^\varepsilon\rangle := \frac{\Pi_{\mathbb{Z}(\varepsilon)} |\text{GKP}_{\kappa,\Delta}\rangle}{\|\Pi_{\mathbb{Z}(\varepsilon)} |\text{GKP}_{\kappa,\Delta}\rangle\|}.$$

Truncation forces each Gaussian peak to vanish outside its strip of width 2ε . The resulting peaks at distinct integers are then *exactly* orthogonal:

Lemma 2.12

For $\varepsilon < 1/2$ and $z, z' \in \mathbb{Z}$: $\langle \chi_\Delta^\varepsilon(z) | \chi_\Delta^\varepsilon(z') \rangle = \delta_{z,z'}$.

Proof. The intervals $[z - \varepsilon, z + \varepsilon]$ and $[z' - \varepsilon, z' + \varepsilon]$ are disjoint when $|z - z'| \geq 1 > 2\varepsilon$, so the product of indicator functions vanishes; for $z = z'$ the inner product is 1 by normalization. See Appendix A.6. $\square$

3. The gates of the algorithm

We now derive the action of every gate used in the algorithm, in order of increasing complexity. Throughout, $x_0 := x \bmod 2$ denotes the least significant bit of an integer x .

The gates are built in three layers. At the bottom are the **primitive gates**: scalar multiplication M_α , displacement e^{-iRP} , qubit-controlled versions of these, and the LSB-extraction gate U^{LSB} . These are the elementary moves available on a CV system with a single qubit. In the middle is the **intermediate gate** V_α : a composite circuit that, in one application, processes a single bit of A's binary expansion and conditionally multiplies B by α . At the top is $V_{a,N,m}$, which chains m copies of V_α together to implement *coherent repeated squaring* — the CV analogue of the modular exponentiation subroutine in Shor's algorithm. Finally, $U_{a,N,m}$ converts $V_{a,N,m}$'s multiplicative action into the additive form needed to mimic Shor's modular-exponentiation unitary $|x\rangle|y\rangle \rightarrow |x\rangle|(y + a^x) \bmod N\rangle$. Figure 5 collects all of these gates—primitive and composite—together with their circuit implementations; we derive each below.

3.1. The LSB gate $U_{A \rightarrow Q}^{\text{LSB}}$

The LSB gate reads out the least significant bit of an integer in oscillator A and writes it into a qubit Q. We need this because the classical *repeated-squaring* algorithm for $a^x \bmod N$ processes the bits of x one at a time: each step looks at one bit of x and, depending on its value, multiplies an accumulator by the next power a^{2^i} or leaves it alone. To do this coherently in superposition, we need a way to peel off one bit of x into a *controllable* register — and that's what the qubit Q does, briefly. The LSB gate is what allows us to use a single qubit as a one-bit scratchpad for the entire algorithm.

The gate has the following circuit:

$$U_{A \rightarrow Q}^{\text{LSB}} := (I_A \otimes H_Q) \circ (\text{ctrl-}e^{i\pi Q_A}) \circ (I_A \otimes H_Q),$$

where H is the Hadamard gate and the qubit is the control of $e^{i\pi Q_A}$.

Lemma 3.1

For $x \in \mathbb{Z}$ and $b \in \{0, 1\}$:

$$U_{A \rightarrow Q}^{\text{LSB}}(|x\rangle_A \otimes |b\rangle_Q) = |x\rangle_A \otimes |b \oplus x_0\rangle_Q.$$

Proof. We track the state. After H_Q :

$$|x\rangle_A \otimes \frac{|0\rangle_Q + (-1)^b |1\rangle_Q}{\sqrt{2}}.$$

The controlled $e^{i\pi Q_A}$ has operator form $I_A \otimes |0\rangle\langle 0|_Q + e^{i\pi Q_A} \otimes |1\rangle\langle 1|_Q$. Applied:

$$\frac{1}{\sqrt{2}} \left(|x\rangle_A |0\rangle_Q + (-1)^b e^{i\pi x} |x\rangle_A |1\rangle_Q \right) = |x\rangle_A \otimes \frac{|0\rangle_Q + (-1)^{b+x_0} |1\rangle_Q}{\sqrt{2}},$$

using $e^{i\pi x} = (-1)^x = (-1)^{x_0}$ for $x \in \mathbb{Z}$. The final H_Q then maps the qubit state to $|b \oplus x_0\rangle_Q$. $\square$

Remark 3.1. *Although the qubit is formally the control of the central gate, information flows from oscillator to qubit by phase kickback: the eigenvalue $(-1)^{x_0}$ produced when $e^{i\pi Q_A}$ acts on $|x\rangle_A$ is converted by the Hadamards into a basis-state flip of the qubit.*

3.2. The qubit-controlled multiplication ctrl- M_α

This gate conditionally multiplies the position of an oscillator by α , depending on the value of a control qubit. It is the primitive that lets us implement *one step* of the repeated-squaring algorithm: given a bit b (extracted by the LSB gate above), if $b = 1$ we want to multiply our running accumulator by a^{2^i} , and if $b = 0$ we want to leave it alone — exactly what ctrl- M_α does with $\alpha = a^{2^i} \bmod N$. The clever circuit construction uses the fact that conjugating squeezing by a 90-degree phase-space rotation turns it into its inverse, so ctrl- $M_{\sqrt{\alpha}} \cdot \text{ctrl-}M_{\sqrt{\alpha}} = \text{ctrl-}M_\alpha$ can be implemented with only four uncontrolled operations sandwiched between phase-space rotations.

Lemma 3.2

The circuit

$$\text{ctrl-}M_\alpha := (\text{ctrl-}e^{i\pi \hat{N}/2}) \circ M_{\sqrt{\alpha}}^\dagger \circ (\text{ctrl-}e^{-i\pi \hat{N}/2}) \circ M_{\sqrt{\alpha}}$$

(reading right to left), where $\hat{N} = (Q^2 + P^2 - 1)/2$ is the number operator, implements

$$\text{ctrl-}M_\alpha(|b\rangle_Q |x\rangle_A) = |b\rangle_Q \left| \alpha^b x \right\rangle_A.$$

Proof. If $b = 0$, both controlled rotations are inactive and $M_{\sqrt{\alpha}}^\dagger M_{\sqrt{\alpha}} = I$. If $b = 1$, all four gates are active; using the rotation identity $RS(z)R^\dagger = S(-z)$ for $R = e^{-i\pi \hat{N}/2}$ (proven in Appendix A.4), the composition reduces to $M_{\sqrt{\alpha}} \cdot M_{\sqrt{\alpha}} = M_\alpha$. $\square$

3.3. The intermediate gate V_α

The gate V_α is the workhorse of the algorithm. In one application, it processes a *single bit* of the input integer x stored in mode A: it reads off the least significant bit $x_0 = x \bmod 2$, and if $x_0 = 1$ it multiplies mode B by α (otherwise it leaves B alone). It then shifts A right by one bit, so the next V_α will see the next bit of x . The gate also moves x_0 into mode C (where it accumulates as A's bits are consumed), and crucially returns the qubit to $|0\rangle$.

This single-bit processing is the building block of *repeated squaring*, the classical algorithm for computing $a^x = \prod_i (a^{2^i})^{x_i}$ one bit at a time. Chaining m copies of V_α with $\alpha = a^{2^0}, a^{2^1}, \dots, a^{2^{m-1}}$ (each reduced mod N) implements the full $a^x \bmod N$ computation coherently in superposition. This is the role of $V_{a,N,m}$, the next layer up.

The V_α gate in seven steps: The circuit definition below may look arbitrary but in fact each step solves a specific design problem.

1. M_2 on C makes room: it scales C's contents up by 2, freeing the LSB slot so we can write x_0 there in step 5.

2. $U_{A \rightarrow Q}^{\text{LSB}}$ extracts x_0 into the qubit. The qubit now holds the single bit that controls the rest of the gate.
3. **Controlled** e^{iP_A} subtracts x_0 from A. After this step, A holds an even integer, so dividing by 2 in step 7 is exact.
4. **Controlled** M_α **on B** performs the actual computational step: multiply B by α if and only if $x_0 = 1$, i.e. compute $\alpha^{x_0} \cdot y$.
5. **Controlled** e^{-iP_C} adds x_0 to C, putting the bit into the LSB slot we freed in step 1.
6. $U_{C \rightarrow Q}^{\text{LSB}}$ uncomputes the qubit: since the LSB of $2z + x_0$ is exactly x_0 , this reads x_0 back into the qubit and XORs it with the qubit's current value, resetting it to $|0\rangle$. This is essential: we cannot leave junk in the qubit, or the next V_α will fail.
7. $M_{1/2}$ **on A** divides A by 2. Since A is even after step 3, this gives the exact integer $(x - x_0)/2 = \sum_{j \geq 1} x_j 2^{j-1}$ — A has been shifted right by one bit. The LSB of A is now what was previously the next bit x_1 .

The seven steps are tightly choreographed between each other, so that each one prepares the conditions for the next, and the symmetric LSB pattern of extracting then uncomputing around the central computation ensures the qubit is always returned clean, i.e. to state 0.

Definition 3.3: V_α

For $\alpha > 0$, V_α is the circuit

$$V_\alpha := M_{1/2}^{(A)} \circ U_{C \rightarrow Q}^{\text{LSB}} \circ (\text{ctrl-}e^{-iP_C}) \circ (\text{ctrl-}M_\alpha^{(B)}) \circ (\text{ctrl-}e^{iP_A}) \circ U_{A \rightarrow Q}^{\text{LSB}} \circ M_2^{(C)},$$

acting on $\mathcal{H} = L^2(\mathbb{R})_A \otimes L^2(\mathbb{R})_B \otimes L^2(\mathbb{R})_C \otimes \mathbb{C}_Q^2$.

Theorem 3.4: Action of V_α

For $x \in \mathbb{Z}$, $y \in \mathbb{R}$, $z \in \mathbb{Z}$:

$$V_\alpha(|x\rangle_A |y\rangle_B |z\rangle_C |0\rangle_Q) = \left| \frac{x-x_0}{2} \right\rangle_A |\alpha^{x_0} y\rangle_B |2z + x_0\rangle_C |0\rangle_Q.$$

Proof. We track $|\Psi_0\rangle = |x\rangle |y\rangle |z\rangle |0\rangle$ through each step.

Step 1 ($M_2^{(C)}$): $|\Psi_1\rangle = |x\rangle |y\rangle |2z\rangle |0\rangle$.

Step 2 ($U_{A \rightarrow Q}^{\text{LSB}}$, Lemma 3.1): $|\Psi_2\rangle = |x\rangle |y\rangle |2z\rangle |x_0\rangle$.

Step 3 ($\text{ctrl-}e^{iP_A}$): The operator is $I_A \otimes |0\rangle\langle 0|_Q + e^{iP_A} \otimes |1\rangle\langle 1|_Q$. With qubit in $|x_0\rangle$ and using $e^{iP} |x\rangle = |x-1\rangle$, the A register becomes $|x-x_0\rangle$ (note this is even):

$$|\Psi_3\rangle = |x-x_0\rangle |y\rangle |2z\rangle |x_0\rangle.$$

Step 4 ($\text{ctrl-}M_\alpha^{(B)}$): B becomes $|\alpha^{x_0} y\rangle$.

Step 5 ($\text{ctrl-}e^{-iP_C}$): C becomes $|2z + x_0\rangle$ (translation by $+x_0$).

Step 6 ($U_{C \rightarrow Q}^{\text{LSB}}$): The LSB of $2z + x_0$ is x_0 (since $2z$ is even), so the qubit becomes $|x_0 \oplus x_0\rangle = |0\rangle$.

Step 7 ($M_{1/2}^{(A)}$): Since $x - x_0$ is even, $(x - x_0)/2 \in \mathbb{Z}$, giving $|(x - x_0)/2\rangle$. □

The gate V_α *peels* the LSB of A, *conditionally multiplies* B by α , *appends* the bit to C, and *right-shifts* A. The qubit is rented as a single-bit register and returned clean.

Corollary 3.5

$V_1^\dagger$ acts as $(|x\rangle|y\rangle|z\rangle|0\rangle) \mapsto (|2x + z_0\rangle|y\rangle|(z - z_0)/2\rangle|0\rangle)$, where $z_0 = z \bmod 2$.

Proof. Set $\alpha = 1$ in Theorem 3.4: V_1 maps $(x, z) \rightarrow ((x - x_0)/2, 2z + x_0)$ (B untouched). Invert this map directly. $\square$

3.4. The pseudomodular power and $V_{a,N,m}$

The gate $V_{a,N,m}$ multiplies the position of mode B by $a^x \bmod N$, where x is the integer stored in mode A. It is the CV analogue of the *modular exponentiation* subroutine of Shor's algorithm — but realized as a multiplicative, not additive, operation. Recall that Shor's algorithm needs the unitary $|x\rangle|y\rangle \rightarrow |x\rangle|(y + a^x) \bmod N\rangle$; we will obtain this additive form in the next subsection (as $U_{a,N,m}$) by a conjugation trick built on $V_{a,N,m}$.

The classical idea $V_{a,N,m}$ implements is *repeated squaring*: rather than computing a^x by x successive multiplications by a (which would take exponential time), one writes $x = \sum_i x_i 2^i$ in binary and computes

$$a^x = \prod_{i: x_i=1} a^{2^i},$$

where the powers $a^{2^i} = (a^{2^{i-1}})^2$ are precomputed by m squarings. This requires only $O(m) = O(\log x)$ multiplications.

In the CV setting, this is implemented as follows: each V_α with $\alpha = a^{2^i} \bmod N$ peels the bit x_i from A and conditionally multiplies B by $\alpha = a^{2^i} \bmod N$. After m such steps, B holds $\prod_i (a^{2^i} \bmod N)^{x_i}$, which equals $a^x \bmod N$ for $x < 2^m$ (this is the "pseudomodular power" defined below). But the bit-peeling has garbled modes A and C; a cleanup pass of $V_1^\dagger$'s (which leave B fixed since $\alpha = 1$ implies no multiplication) reverses this garbling.

Definition 3.6: Pseudomodular power

For $a, N, m \in \mathbb{N}$ and $x \in \mathbb{N}_0$ with binary expansion $x = \sum_{j \geq 0} x_j 2^j$:

$$f_{a,N,m}(x) := \prod_{i=0}^{m-1} (a^{2^i} \bmod N)^{x_i}.$$

For $x \in \{0, \dots, 2^m - 1\}$ we have $f_{a,N,m}(x) \equiv a^x \pmod{N}$, but the product is computed as a real number, not reduced mod N . This is the classical *repeated-squaring* algorithm executed coherently. A worked example with $a = 3$, $N = 5$, $m = 4$, $x = 11$ gives $f_{3,5,4}(11) = 3 \cdot 4 \cdot 1 \cdot 1 = 12$, while $3^{11} \equiv 2 \pmod{5}$ and $12 \bmod 5 = 2$.

Definition 3.7: $V_{a,N,m}$

$$V_{a,N,m} := (V_1^\dagger)^m \circ V_{a^{2^{m-1}} \bmod N} \circ \dots \circ V_{a^{2^0} \bmod N}.$$

Theorem 3.8: Action of $V_{a,N,m}$

For $x \in \mathbb{N}_0$, $y \in \mathbb{R}$, $z \in \mathbb{N}_0$:

$$V_{a,N,m}(|x\rangle_A |y\rangle_B |z\rangle_C |0\rangle_Q) = |x\rangle_A |f_{a,N,m}(x) \cdot y\rangle_B |z\rangle_C |0\rangle_Q.$$

Proof sketch. The forward pass of m V_α -gates, by induction on j , leaves the system in state

$$\left| \frac{x - \sum_{i < j} x_i 2^i}{2^j} \right\rangle_A \left| \prod_{i < j} (a^{2^i} \bmod N)^{x_i} \cdot y \right\rangle_B \left| 2^j z + \sum_{i < j} 2^{j-1-i} x_i \right\rangle_C |0\rangle_Q$$

after step j . After $j = m$, mode B holds the answer $f_{a,N,m}(x) \cdot y$, but A and C are garbled. The cleanup pass $(V_1^\dagger)^m$ undoes the bit-transfer, restoring A and C exactly. Full induction in Appendix A.7. $\square$

3.5. The controlled translation $U_{a,N,m}$

This is the direct CV analogue of Shor's modular exponentiation unitary $U_{Z_N,a} : |x\rangle |y\rangle \rightarrow |x\rangle |(y + a^x) \bmod N\rangle$. The gate $V_{a,N,m}$ already computes $a^x \bmod N$, but it produces a *multiplicative* effect ($y \rightarrow f(x) \cdot y$); we need an *additive* one ($y \rightarrow y + f(x)$) for two reasons. First, the addition is what allows the GKP lattice trick of mode B to enforce modularity (translations by integer multiples of N leave the lattice approximately invariant). Second, this additive form is what creates the entanglement structure $\sum_x |x\rangle |f(x) \bmod N\rangle$ that period-finding depends on.

The conjugation trick. How do we convert a multiplication into an addition? The key identity is purely algebraic: for any nonzero $f(x)$,

$$y + f(x) = f(x) \cdot \left(1 + \frac{y}{f(x)} \right).$$

Reading right to left, this says: *divide y by $f(x)$, add 1, multiply back by $f(x)$* . The first and third operations are exactly $V_{a,N,m}^\dagger$ and $V_{a,N,m}$. The middle operation is just a fixed translation e^{-iP} of B by $+1$, which is a primitive gate. So the entire data-dependent translation $y \rightarrow y + f(x)$ is implemented by

$$U_{a,N,m} = V_{a,N,m} \circ e^{-iP_B} \circ V_{a,N,m}^\dagger,$$

a conjugation. This is a very general technique: it converts any multiplicative quantum operation into the corresponding additive one by sandwiching a fixed-constant translation. It is the moment in the algorithm where the multiplicative computation of $V_{a,N,m}$ is transformed into the form needed to interact with mode B's lattice.

Definition 3.9: $U_{a,N,m}$

$$U_{a,N,m} := V_{a,N,m} \circ (I_A \otimes e^{-iP_B} \otimes I_C \otimes I_Q) \circ V_{a,N,m}^\dagger.$$

Theorem 3.10: Action of $U_{a,N,m}$

For $x \in \mathbb{N}_0$, $y \in \mathbb{R}$:

$$U_{a,N,m}(|x\rangle_A |y\rangle_B |0\rangle_C |0\rangle_Q) = |x\rangle_A |y + f_{a,N,m}(x)\rangle_B |0\rangle_C |0\rangle_Q.$$

Proof. Direct calculation, using $V_{a,N,m}^\dagger$ to divide B by $f_{a,N,m}(x)$, then e^{-iP_B} to add 1, then $V_{a,N,m}$ to multiply back by $f_{a,N,m}(x)$:

$$y \xrightarrow{V^\dagger} \frac{y}{f_{a,N,m}(x)} \xrightarrow{e^{-iP_B}} 1 + \frac{y}{f_{a,N,m}(x)} \xrightarrow{V} f_{a,N,m}(x) + y. \quad \square$$

This is the continuous-variable analogue of Shor's modular exponentiation unitary. The conjugation trick converts a fixed unit displacement into a data-dependent displacement of magnitude $f_{a,N,m}(x)$.

3.6. Summary of gates

The full circuits of the gates are located in the Appendix E [?].

Gate	Action on position eigenstates	Cost
M_α	$ x\rangle \rightarrow \alpha x\rangle$	$O(\log \alpha)$
e^{-iRP}	$ x\rangle \rightarrow x + R\rangle$	$O(\log R)$
ctrl- M_α	$ b\rangle x\rangle \rightarrow b\rangle \alpha^b x\rangle$	$O(\log \alpha)$
$U_{A \rightarrow Q}^{\text{LSB}}$	$ x\rangle b\rangle \rightarrow x\rangle b \oplus x_0\rangle$	$O(1)$
V_α	$ x, y, z, 0\rangle \rightarrow \left \frac{x-x_0}{2}, \alpha^{x_0} y, 2z + x_0, 0 \right\rangle$	$O(\log \alpha)$
$V_{a,N,m}$	$ x, y, z, 0\rangle \rightarrow x, f_{a,N,m}(x)y, z, 0\rangle$	$O(m \log N)$
$U_{a,N,m}$	$ x, y, 0, 0\rangle \rightarrow x, y + f_{a,N,m}(x), 0, 0\rangle$	$O(m \log N)$

4. The idealized algorithm

We first analyze an idealized version of the algorithm in which mode A is initialized in the (non-normalizable) Dirac comb $|\text{GKP}\rangle \propto \sum_{x \in \mathbb{Z}} |x\rangle$ and mode C in the position eigenstate $|0\rangle$. The qualitative behavior is the same as in the physical algorithm, and many features of the latter are clearest in this clean limit.

Intuition. Before diving into the calculation, here is the high-level story of what the algorithm does, step by step:

1. **Prepare a uniform superposition.** Mode A enters $|\text{GKP}\rangle = \sum_x |x\rangle$ — the CV analogue of $\sum_x |x\rangle$ on the input register in Shor.
2. **Prepare the lattice register.** Mode B enters $M_N |\text{GKP}\rangle = \sum_y |yN\rangle$ — a lattice with spacing N . This is where the "mod N " structure comes from: anything we add to B will be implicitly considered modulo N because translations by N leave the lattice invariant.
3. **Compute $a^x \bmod N$ into B.** Applying $U_{a,N,m}$ gives the entangled state $\sum_x |x\rangle |a^x \bmod N + N\mathbb{Z}\rangle$. Mode A and mode B are now correlated: knowing the residue in B reveals which coset $\{x : a^x \equiv k\}$ we are in.
4. **Observe (or ignore) B.** Whether we actually measure B or not, tracing it out leaves A in a mixed state over period- r Dirac combs $\sum_\ell |j + \ell r\rangle$, one for each residue class k . The unknown random shift j encodes which class.

5. **P -measure A.** A homodyne P -measurement on a period- r comb produces a sample $w = n/r$ for some random $n \in \mathbb{Z}$. The unknown shift j becomes a phase $e^{-2\pi i j w}$ that vanishes under modulus squared — the period r is the only information that survives.
6. **Recover r classically.** The sample $w = n/r$ is processed by continued fractions to recover the denominator r , then N is factored via $\gcd(a^{r/2} \pm 1, N)$.

The remarkable feature is that the algorithm *never explicitly computes a discrete Fourier transform*. The P -quadrature measurement is a continuous Fourier transform implemented at the level of physical hardware, and that is exactly what is needed to extract the period of a Dirac comb.

4.1. Circuit and initial state

The idealized algorithm $W_{a,N}$ has the structure

$$W_{a,N} = [P\text{-measurement on A}] \circ U_{a,N,m} \circ (e^{-iRP_A} \otimes M_N^{(B)} \otimes I_C \otimes I_Q)$$

applied to

$$|\Psi_0\rangle = |\text{GKP}\rangle_A \otimes |\text{GKP}\rangle_B \otimes |0\rangle_C \otimes |0\rangle_Q.$$

This circuit is shown in Figure 3.

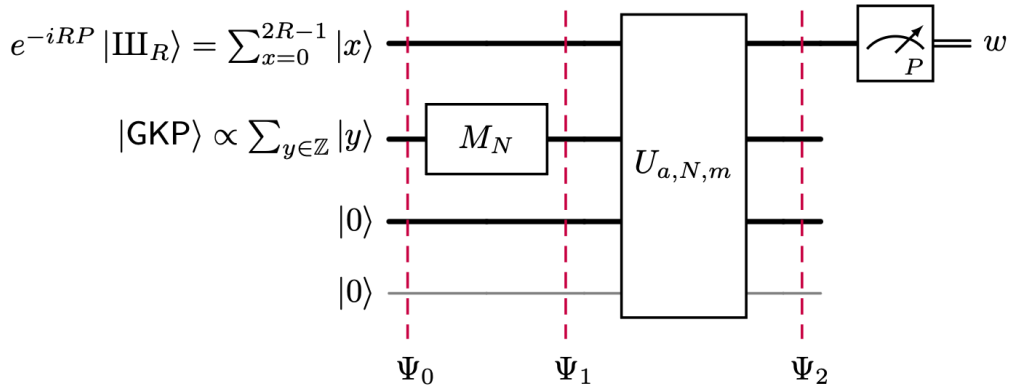


Figure 3: The idealized circuit $W_{a,N}$ acting on non-normalizable comb states. The dashed cuts mark the intermediate states Ψ_0 , Ψ_1 , Ψ_2 tracked in the state evolution below. Reproduced from [1].

4.2. State evolution

After preparation. The translation e^{-iRP} shifts A:

$$e^{-iRP} |\text{GKP}\rangle \propto \sum_{x \in \mathbb{Z}} |x + R\rangle = \sum_{x \in \mathbb{Z}} |x\rangle,$$

formally invariant under re-indexing; in the physical version this shift centers the comb on positive integers $\{0, \dots, 2R - 1\}$. The scaling M_N transforms $|\text{GKP}\rangle_B$ into a lattice with spacing N :

$$M_N |\text{GKP}\rangle \propto \sum_{y \in \mathbb{Z}} |yN\rangle.$$

The prepared state is therefore

$$|\Psi_1\rangle \propto \sum_{x=0}^{2R-1} \sum_{y \in \mathbb{Z}} |x\rangle_A |yN\rangle_B |0\rangle_C |0\rangle_Q.$$

After $U_{a,N,m}$. Apply Theorem 3.10 term by term:

$$|\Psi_2\rangle \propto \sum_{x=0}^{2R-1} \sum_{y \in \mathbb{Z}} |x\rangle_A |yN + f_{a,N,m}(x)\rangle_B |0\rangle_C |0\rangle_Q.$$

Two crucial observations:

1. Provided $2^m > 2R$, the relation $f_{a,N,m}(x) \equiv a^x \pmod{N}$ holds for $x \in \{0, \dots, 2R-1\}$. With Table I parameters this is automatic.
2. Mode C and the qubit decouple from A and B (they end in $|0\rangle|0\rangle$) and may be ignored hereafter.

4.3. Reorganizing by remainder class

Define

$$\text{Rem}_{a,N} := \{a^x \bmod N : x \in \{0, \dots, 2R-1\}\}, \quad S_k := \{x : f_{a,N,m}(x) \equiv k \pmod{N}\}.$$

The sets $\{S_k\}_{k \in \text{Rem}_{a,N}}$ partition $\{0, \dots, 2R-1\}$. Rewriting:

$$|\Psi_2\rangle_{AB} \propto \sum_{k \in \text{Rem}_{a,N}} \sum_{x \in S_k} \sum_{y \in \mathbb{Z}} |x\rangle_A |yN + k\rangle_B.$$

Structure of S_k . Let $r = r(a)$ denote the order of a in $\mathbb{Z}_N^*$ (the period of $x \mapsto a^x \bmod N$). Then $a^x \equiv k \pmod{N}$ iff $x \equiv j_k \pmod{r}$ for some unique $j_k \in \{0, \dots, r-1\}$ (with $a^{j_k} \equiv k$). Hence

$$S_k = \{j_k, j_k + r, j_k + 2r, \dots\} \cap \{0, \dots, 2R-1\}.$$

4.4. Measurement of mode B

A Q -quadrature measurement on B yields a value of the form $w_B = \ell N + k$ for some $\ell \in \mathbb{Z}$ and $k \in \text{Rem}_{a,N}$. The measurement reveals $k = w_B \bmod N$ but not ℓ . Conditioned on the outcome having residue k , mode A collapses to the period- r comb

$$|\Psi_A^{(k)}\rangle \propto \sum_{\ell \in \mathbb{Z}} |j_k + \ell r\rangle_A.$$

The same conclusion holds, by linearity, even if we do not actually measure B: tracing out B leaves A in the mixed state $\rho_A = \frac{1}{r} \sum_k |\Psi_A^{(k)}\rangle\langle\Psi_A^{(k)}|$.

4.5. P -quadrature measurement of A

A homodyne P -measurement is equivalent to a Fourier transform followed by a Q -measurement, giving outcomes $w \in \mathbb{R}$ with density

$$p(w) \propto \left| \widehat{\Psi_A^{(k)}}(w) \right|^2.$$

The Fourier transform of a Dirac comb of spacing r shifted by j_k is (see Appendix C.1)

$$\sum_{\ell \in \mathbb{Z}} \widehat{\delta(x - j_k - \ell r)}(w) = \frac{1}{r} e^{-2\pi i j_k w} \sum_{n \in \mathbb{Z}} \delta(w - n/r).$$

Taking the modulus squared kills the phase, so the output distribution is

Output distribution of the idealized algorithm:

$$p(w) = \frac{1}{r} \sum_{n \in \mathbb{Z}} \delta\left(w - \frac{n}{r}\right).$$

The distribution is uniform on the lattice $\frac{1}{r}\mathbb{Z}$ and *independent of j_k* . The unknown shift j_k , which encodes which residue class we landed in, is washed out by the modulus squared. Information about the period r alone survives.

4.6. Classical post-processing

The measurement gives $w = n/r$ for some unknown $n \in \mathbb{Z}$. The continued fraction algorithm (Appendix C.3) produces a sequence of convergents $p_0/q_0, p_1/q_1, \dots$ approximating w . For each convergent, test the candidate q_k for the period: reject if $q_k \geq N$ or if $a^{q_k} \not\equiv 1 \pmod{N}$, accept the smallest passing q_k .

This succeeds whenever $\gcd(n, r) = 1$, which has probability $\phi(r)/r \geq \Omega(1/\log \log r)$. Repeating $O(\log \log N)$ times succeeds with probability at least $1 - 1/e$.

Once r is known and even with $a^{r/2} \not\equiv -1 \pmod{N}$, the factors are obtained via

$$\gcd(a^{r/2} - 1, N), \quad \gcd(a^{r/2} + 1, N).$$

A random choice of a satisfies the conditions with probability $\geq 1/2$ [2].

Worked example. Take $N = 15$, $a = 2$ (true period $r = 4$). Suppose $w = 0.749 \approx 3/4$. In nested-fraction form:

$$0.749 = \frac{749}{1000} = \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \frac{1}{247/4}}}}.$$

The convergents and tests:

k	p_k/q_k	$a^{q_k} \bmod N$	Verdict
0	0/1	2	reject
1	1/1	2	reject
2	2/3	8	reject
3	3/4	1	accept: $r = 4$

Then $\gcd(2^2 - 1, 15) = 3$ and $\gcd(2^2 + 1, 15) = 5$, recovering $15 = 3 \times 5$.

5. The physical algorithm

The idealized algorithm uses non-normalizable comb states and exact integer arithmetic; both are unphysical. The physical algorithm $Q_{a,N}$ replaces the Dirac combs by finitely-squeezed approximate GKP states. The same circuit then computes, with explicit error bounds.

What changes and what doesn't. The circuit is structurally the same — same gates, same composition, same measurement. What changes is the input states: instead of non-normalizable combs, we use finitely-squeezed GKP states where each "peak" is a tiny Gaussian and the comb is multiplied by a wide Gaussian envelope. Four sources of error are introduced:

1. **Peaks are Gaussians, not deltas.** Each "position eigenstate" $|x\rangle$ becomes $|\chi_\Delta(x)\rangle$, a Gaussian of width Δ around x . This breaks the exact LSB extraction (the gate gives a phase $e^{i\pi y}$ for y near integer, not exactly equal).
2. **Combs are finite.** A Gaussian envelope of width $1/\kappa$ truncates the sum to roughly $1/\kappa$ peaks, instead of all of $\mathbb{Z}$. The sum over x in the analysis becomes bounded.
3. **Modular invariance is approximate.** The lattice state $M_N |\text{GKP}\rangle$ is not *exactly* invariant under translations by N , because the Gaussian envelope changes value when shifted by N . The error is small when $\kappa \cdot N \ll 1$.
4. $U_{a,N,m}$ **is approximate.** The LSB gate inside each V_α accumulates a small phase error per step; the total error scales linearly with the number of V_α applications and with the peak width Δ .

The art of the proof is to bound each of these four errors and show that, with appropriately chosen parameters (Table I below), all four are exponentially small in n . The overall trace distance between the actual output state and an idealized "reference state" is bounded by $364 \cdot 2^{-2n}$, which is tiny compared to the measurement signal ($\Omega(1/r) \geq \Omega(2^{-n})$). Thus the physical algorithm succeeds with essentially the same probability as the idealized one.

5.1. Parameters

Let $n := \lceil \log_2 N \rceil$. The paper specifies (Table I):

Parameter	Value
m	$17n$
R	2^{17n-1}
κ_A, Δ_A	2^{-16n}
κ_B, Δ_B	2^{-18n^2}
Δ_C	2^{-50n}

The mode-B parameters scale as $2^{-\Theta(n^2)}$, which is the source of the algorithm's enormous energy cost (Section 6).

5.2. The circuit $Q_{a,N}$

The same unitary as before is applied to the input state

$$|\Psi_0\rangle = |\text{GKP}_{\kappa_A, \Delta_A}\rangle_A \otimes |\text{GKP}_{\kappa_B, \Delta_B}\rangle_B \otimes |\text{vac}\rangle_C \otimes |0\rangle_Q,$$

followed by squeezing of C to $|\Psi_{\Delta_C}\rangle_C := M_{\Delta_C} |\text{vac}\rangle_C$, scaling of B by M_N , and shifting of A by e^{-iRP_A} . We then apply $U_{a,N,m}$ and measure A in the P -quadrature. The full circuit is shown in Figure 4.

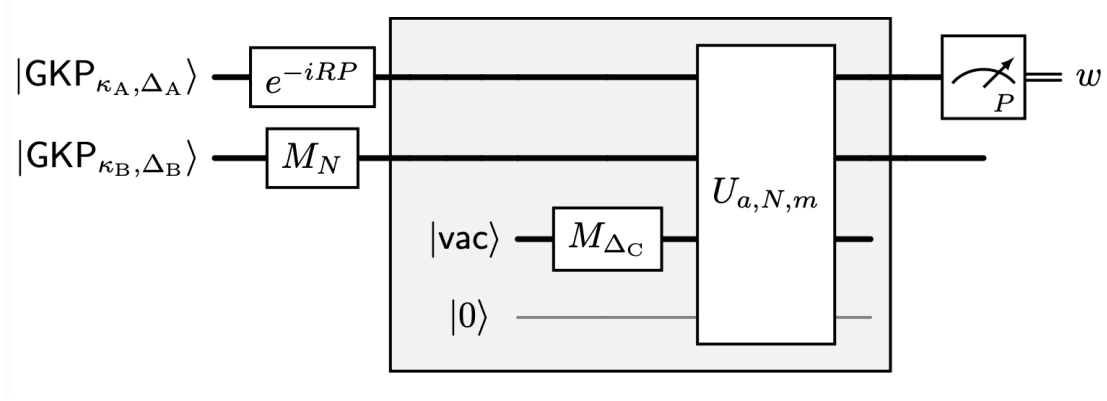


Figure 4: The physical circuit $Q_{a,N}$: finitely-squeezed GKP inputs on modes A and B and a squeezed vacuum on C, the preparation steps (e^{-iRP} , M_N , M_{Δ_C}), the controlled translation $U_{a,N,m}$, and the homodyne P -measurement returning the outcome w . Reproduced from [1].

5.3. Preparation state

Mode A. Shifting the envelope to R :

$$e^{-iRP_A} |\text{GKP}_{\kappa_A, \Delta_A}\rangle = C_{\kappa_A, \Delta_A} \sum_{z \in \mathbb{Z}} \eta_{\kappa_A}(z - R) |\chi_{\Delta_A}(z)\rangle.$$

The Gaussian envelope $\eta_{\kappa_A}(\cdot - R)$ is centered at R with width $1/\kappa_A$, so the support is concentrated on $z \in \{0, \dots, 2R - 1\}$.

Mode B. Scaling by N :

Lemma 5.1

$$M_N |\text{GKP}_{\kappa_B, \Delta_B}\rangle \propto \sum_{y \in N\mathbb{Z}} \eta_{\kappa_B/N}(y) |\chi_{N\Delta_B}(y)\rangle.$$

Proof. Compute $M_N |\chi_{\Delta_B}(z)\rangle = |\chi_{N\Delta_B}(Nz)\rangle$ at the wavefunction level (Appendix A.5). Change summation variable $y = Nz$ and absorb the constant ratio $\eta_{\kappa_B}(y/N)/\eta_{\kappa_B/N}(y) = \sqrt{N}$ into the normalization. $\square$

The resulting lattice has spacing N , peak width $N\Delta_B$, and envelope width N/κ_B . Crucially, since $\kappa_B \cdot N = 2^{-18n^2+n} \ll 1$, the envelope is much wider than N , so the state is *approximately invariant* under translations by integer multiples of N (quantified below).

Mode C. $M_{\Delta_C} |\text{vac}\rangle = |\chi_{\Delta_C}(0)\rangle =: |\Psi_{\Delta_C}\rangle$, a Gaussian of width Δ_C centered at 0.

5.4. Approximate action of $U_{a,N,m}$

The exact action $|x\rangle |y\rangle |0\rangle |0\rangle \rightarrow |x\rangle |y + f_{a,N,m}(x)\rangle |0\rangle |0\rangle$ of Theorem 3.10 requires sharp integer inputs in A and C. The physical inputs are Gaussian-smeared, so the gate is exact only approximately. The error is controlled by:

Lemma 5.2: Approximate function evaluation; Lemma 5.10 of [1]

Let Ψ_A be supported within $\varepsilon_A \in (0, 1/4)$ of $x \in \mathbb{N}_0 \cap \{0, \dots, 2^m - 1\}$, and Ψ_C supported within $\varepsilon_C \in (0, 2^{-(m+2)})$ of 0. Then

$$\|U_{a,N,m}(\Psi_A \otimes \Psi_B \otimes \Psi_C \otimes |0\rangle) - \Psi_A \otimes e^{-if_{a,N,m}(x)P_B} \Psi_B \otimes \Psi_C \otimes |0\rangle\| \leq 176 \varepsilon_A^{1/4} + 176 (2^m \varepsilon_C)^{1/4}.$$

The proof is in Appendix B.2. The error is governed by the fourth root $\varepsilon^{1/4}$, by the following mechanism. A single LSB gate incurs an *overlap* deficit that is linear in the support width, $|1 - \langle \text{ideal} | \text{actual} \rangle| \leq \pi \varepsilon / 2$ (Lemma 3.1; see Appendix B.1). What controls $\|\Psi_a - \Phi_a\|_1$, however, is the *operator* error of the composite unitary acting on a superposition of peaks of both parities: there the cross-terms between the correct and incorrect qubit outcomes are bounded by $\sqrt{2\varepsilon}$ (proof of Lemma 5.3 of [1]), and passing from this overlap deficit to a vector-norm distance contributes a further square root. Hence each LSB gate carries a function-evaluation error $6\varepsilon^{1/4}$ (Lemma 5.3 of [1]); the two gates inside each V_α give $\delta_{V_\alpha} = 7\varepsilon_A^{1/4} + 7(2\varepsilon_C)^{1/4}$ (Lemma 5.5), chaining m of them with the cleanup pass gives $\delta_{V_{a,N,m}} = 88\varepsilon_A^{1/4} + 88(2^m \varepsilon_C)^{1/4}$ (Lemma 5.7), and the conjugation $U_{a,N,m} = V_{a,N,m} e^{-iP_B} V_{a,N,m}^\dagger$ doubles this to the constant 176 of the lemma (Lemma 5.10).

5.5. Approximate invariance of the lattice state

Lemma 5.3: Approximate translation-invariance

For $\ell \in \mathbb{Z}$,

$$\langle M_N \text{GKP}_{\kappa_B, \Delta_B} | e^{-i\ell N P_B} | M_N \text{GKP}_{\kappa_B, \Delta_B} \rangle \approx e^{-(\kappa_B \ell)^2 / 4}.$$

Proof idea. By Lemma 5.1, $e^{-i\ell N P_B}$ shifts each peak from y to $y + \ell N$ (a lattice symmetry), but changes the envelope from $\eta_{\kappa_B/N}(y)$ to $\eta_{\kappa_B/N}(y - \ell N)$. The overlap of the two envelopes is computed by Gaussian integration. Full derivation in Appendix B.3. $\square$

Since $f_{a,N,m}(z) - (a^z \bmod N) \in N\mathbb{Z}$ with quotient at most N^{m-1} , the worst-case ℓ satisfies $|\kappa_B \ell| \leq 2^{-18n^2} \cdot 2^{n(17n-1)} = 2^{-n^2-n}$ (using $m = 17n$), so the inner product is $\approx 1 - 2^{-2n^2-2n-2}$, an exponentially small error in n^2 .

5.6. The reference state and main approximation theorem

Why introduce a reference state? The actual output Ψ_a of the circuit accumulates errors from all four sources listed in the section opening. Analyzing the measurement distribution of Ψ_a directly would require tracking all these errors through Fourier analysis simultaneously — a notational and conceptual nightmare. Instead, we introduce an idealized *reference state* Φ_a that is mathematically clean (acts as if all approximations were exact) and prove two things separately: (i) Ψ_a is close to Φ_a in trace distance, and (ii) Φ_a has a measurement distribution with the right structure for factoring. The triangle inequality then guarantees that the actual measurement distribution also has the right structure, up to a tiny correction. This “introduce an idealized intermediate object” trick is a standard technique in quantum proofs; what makes it work here is that the error bounds are exponentially small in n , while the signal we care about is only polynomially small.

Definition 5.4: Reference state

With $\varepsilon_B = \sqrt{\Delta_B}$ and $\varepsilon_C = \sqrt{\Delta_C}$:

$$|\Phi_a\rangle := c_{\Phi_a} \sum_{z \in \mathbb{Z}} \eta_{\kappa_A}(z - R) |\chi_{\Delta_A}(z)\rangle_A \\ \otimes e^{-i(a^z \bmod N)P_B} M_N \left| \text{GKP}_{\kappa_B, \Delta_B}^{\varepsilon_B} \right\rangle_B \otimes \left| \Psi_{\Delta_C}^{\varepsilon_C} \right\rangle_C \otimes |0\rangle_Q.$$

The reference state is the output one would obtain if all four sources of approximation (LSB rounding, lattice translation, peak truncation, envelope truncation) were exact. It is not produced by the circuit, but lives close to what the circuit produces:

Proposition 5.5: Proposition 2 of [1]

For $N \geq 8$ and parameters as in Table I, the actual output Ψ_a of $Q_{a,N}$ before measurement satisfies

$$\|\Psi_a - \Phi_a\|_1 \leq 364 \cdot 2^{-2n}.$$

The proof (Appendix B.4) bounds five contributions. The dominant one is the approximate function evaluation of $U_{a,N,m}$ (Lemma 5.2), which contributes $352 \cdot 2^{-2n}$; truncation of the input states contributes $9 \cdot 2^{-2n}$, and the three remaining steps each contribute 2^{-2n} , for a total of $364 \cdot 2^{-2n}$.

5.7. The measurement distribution

Decompose Φ_a by residue class modulo r : writing $z = jr + k$ with $j \in \mathbb{Z}$, $k \in \{0, \dots, r-1\}$, and using $a^z \equiv a^k \pmod{N}$,

$$|\Phi_a\rangle = c_{\Phi_a} \sum_{k=0}^{r-1} \underbrace{\left(\sum_{j \in \mathbb{Z}} \eta_{\kappa_A}(jr + k - R) |\chi_{\Delta_A}(jr + k)\rangle_A \right)}_{=: |\Phi_a^{(k)}\rangle_A} \\ \otimes \underbrace{e^{-i(a^k \bmod N)P_B} M_N \left| \text{GKP}_{\kappa_B, \Delta_B}^{\varepsilon_B} \right\rangle_B}_{=: |B_k\rangle} \otimes |\dots\rangle_C |0\rangle_Q.$$

The states $\{|B_k\rangle\}$ are approximately orthogonal because they are concentrated near distinct cosets $a^k + N\mathbb{Z}$. Tracing out B (and the inert C, Q):

$$\rho_A \approx \frac{1}{r} \sum_{k=0}^{r-1} \left| \Phi_a^{(k)} \right\rangle \left\langle \Phi_a^{(k)} \right|.$$

Each $|\Phi_a^{(k)}\rangle$ is an approximate GKP-like state with spacing r , envelope width $1/\kappa_A$ centered at R , and peak width Δ_A . Its Fourier transform has spacing $1/r$ and a phase $e^{-2\pi i k w}$; the phase drops out of the modulus squared. The P -quadrature measurement distribution of Φ_a is therefore

$$p_{\Phi_a}(w) \approx \left| \widehat{\Phi_a^{(0)}}(w) \right|^2,$$

a smeared comb concentrated on $\frac{1}{r}\mathbb{Z}$ with peak width κ_A .

5.8. Suitability and the main theorem

Why “suitability”. In Shor’s algorithm, the QFT produces a discrete distribution over $\{0, 1/q, 2/q, \dots, (q-1)/q\}$ for some $q > N^2$, with peaks concentrated near $\{d/r : d = 0, 1, \dots, r-1\}$. The classical post-processing succeeds precisely when the sample w is close enough to one of these d/r values, and the value d is coprime to r . The notion of *suitability* below is the CV analogue of this concentration property: it says that the continuous distribution p_a puts at least $\Omega(1/r)$ probability mass near each of the r lattice points d/r . Once we know suitability holds, the same post-processing as in Shor (continued fractions) will succeed.

Definition 5.6: Suitable family

Let $q = \min\{2^t : 2^t > N^2\}$. For $d \in \{0, \dots, r-1\}$:

$$\Gamma_d(a) := \bigcup_{j \in \mathbb{Z}} \left[j + \frac{d}{r} - \frac{1}{2q}, j + \frac{d}{r} + \frac{1}{2q} \right].$$

A family $\{p_a\}$ of probability densities on $\mathbb{R}$ is *suitable* if for all a and all $d \in \{0, \dots, r(a)-1\}$:

$$\int_{\Gamma_d(a)} p_a(w) dw \geq \Omega(1) \cdot \frac{1}{r(a)}.$$

The set $\Gamma_d(a)$ is the union of small windows of width $1/q < 1/N^2$ around each lattice point d/r . The width $1/q$ is precisely the precision needed for continued fractions to extract the denominator r from a sample $w \approx d/r$.

Proposition 5.7: Proposition 3 of [1]

The family $\{p_{\Phi_a}\}_{a \in \mathbb{Z}_N^*}$ is suitable.

The Gaussian-shaped peaks of $|\widehat{\Phi_a^{(0)}}(w)|^2$ around each d/r are wide enough ($\kappa_A = 2^{-16n}$ in w -space) and tall enough to put $\Omega(1/r)$ mass into each window. Proof in Appendix C.2.

Proposition 5.8: Proposition 1 of [1]

There is a classical algorithm running in time $\text{poly}(n)$ that, given (a, N) and a sample w from a suitable distribution p_a , outputs a non-trivial factor of N with probability $\geq \Omega(1/\log n)$.

The algorithm is the one of Section 4.6: apply continued fractions to w , test each convergent’s denominator as a candidate period r , then compute $\gcd(a^{r/2} \pm 1, N)$.

Theorem 5.9: Theorem 1 of [1]

For $N \geq 8$ and parameters as in Table I, the circuit $Q_{a,N}$ outputs a sample w such that the classical post-processing of Proposition 5.8 recovers a non-trivial factor of N with probability $\geq \Omega(1/\log n)$ per shot. Repeating $O(\log n)$ times succeeds with probability $\geq 1 - 1/e$.

Proof. Combine Propositions 5.5, 5.7, and 5.8. The trace-distance bound $\|\Psi_a - \Phi_a\|_1 \leq 364 \cdot 2^{-2n}$ implies $\|p_{\Psi_a} - p_{\Phi_a}\|_{\text{TV}} \leq 364 \cdot 2^{-2n}$ (data-processing inequality), so $\{p_{\Psi_a}\}$ is also suitable up to a correction of order 2^{-2n} , which is absorbed into the $\Omega(1/r) \geq \Omega(2^{-n})$ bound. $\square$

5.9. The full chain

$$\begin{array}{c}
\underbrace{|\text{GKP}_A\rangle |\text{GKP}_B\rangle |\text{vac}\rangle |0\rangle}_{\Psi_0} \xrightarrow{\text{prep}} |\Psi_1\rangle \xrightarrow{U_{a,N,m}} \underbrace{|\Psi_a\rangle}_{\text{exact}} \overset{\|\cdot\|_1 \leq 364 \cdot 2^{-2n}}{\approx} \underbrace{|\Phi_a\rangle}_{\text{reference}} \\
\\
\underbrace{w \in \Gamma_d(a)}_{\substack{\Omega(1/r) \text{ prob.}}} \xrightarrow{P\text{-meas A}} \xrightarrow{\text{cont. fractions}} r \xrightarrow{\gcd(a^{r/2} \pm 1, N)} \text{factor of } N.
\end{array}$$

6. Energy requirements

The algorithm's elegance comes at a steep cost: the GKP states must be very tightly squeezed, requiring large oscillator energy. We make this quantitative.

6.1. Energy of a squeezed state

For a harmonic oscillator with Hamiltonian $H = \hbar\omega(\hat{N} + \frac{1}{2})$, the vacuum has energy $\frac{1}{2}\hbar\omega$. A squeezed vacuum $S(s)|\text{vac}\rangle$ has, by Lemma 2.3,

$$\begin{aligned}
\langle S(s)|\text{vac}|Q^2|S(s)|\text{vac}\rangle &= e^{-2s}\langle \text{vac}|Q^2|\text{vac}\rangle = \frac{1}{2}e^{-2s}, \\
\langle S(s)|\text{vac}|P^2|S(s)|\text{vac}\rangle &= e^{2s}\langle \text{vac}|P^2|\text{vac}\rangle = \frac{1}{2}e^{2s}.
\end{aligned}$$

(Here $s := \log(1/\Delta)$, so the squeezing $S(s)$ produces a state of position width $\Delta = e^{-s}$.) The mean energy is

$$\langle H \rangle = \frac{\hbar\omega}{2}\langle Q^2 + P^2 \rangle = \frac{\hbar\omega}{4}(e^{-2s} + e^{2s}) = \frac{\hbar\omega}{2} \cosh(2s).$$

For $s \gg 1$ this is approximately $\frac{\hbar\omega}{4}e^{2s} = \frac{\hbar\omega}{4\Delta^2}$.

6.2. Energy of an approximate GKP state

An approximate GKP state $|\text{GKP}_{\kappa,\Delta}\rangle$ has Gaussian structure in both quadratures: peaks of width Δ in Q and (by Fourier duality) of width κ in P . A direct calculation (see Appendix D) gives

$$\langle H \rangle_{\text{GKP}_{\kappa,\Delta}} \approx \frac{\hbar\omega}{4} \left(\frac{1}{\Delta^2} + \frac{1}{\kappa^2} \right).$$

With Table I parameters $\kappa = \Delta$, this simplifies to $\langle H \rangle \approx \frac{\hbar\omega}{2\Delta^2}$.

6.3. Total energy with Table I parameters

Plugging in:

$$\begin{aligned}
\langle H_A \rangle &\approx \frac{\hbar\omega_A}{2} \cdot 2^{32n}, & (\text{using } \kappa_A = \Delta_A = 2^{-16n}) \\
\langle H_B \rangle &\approx \frac{\hbar\omega_B}{2} \cdot 2^{36n^2}, & (\text{using } \kappa_B = \Delta_B = 2^{-18n^2}) \\
\langle H_C \rangle &\approx \frac{\hbar\omega_C}{4} \cdot 2^{100n}. & (\text{squeezed vacuum, } \Delta_C = 2^{-50n})
\end{aligned}$$

Note: mode C does not have a GKP comb structure; it begins as the squeezed vacuum $M_{\Delta_C}|\text{vac}\rangle$, whose energy is $\frac{\hbar\omega_C}{2} \cosh(2 \log(1/\Delta_C)) \approx \frac{\hbar\omega_C}{4\Delta_C^2} = \frac{\hbar\omega_C}{4} \cdot 2^{100n}$.

The total energy is dominated by $\langle H_B \rangle$:

Total energy:

$$\langle H_{\text{tot}} \rangle \approx \frac{\hbar\omega}{2} \cdot 2^{36n^2} \quad (\text{for } \omega_A = \omega_B = \omega_C = \omega).$$

This is exponential in $n^2 = (\log_2 N)^2$.

6.4. Numerical estimates

We take $\hbar\omega = 10^{-24}$ J (a typical microwave-cavity frequency, $\omega/2\pi \sim 10$ GHz; an optical cavity would give $\hbar\omega \sim 10^{-19}$ J, making numbers five orders of magnitude worse).

Two-digit number ($N \leq 100$, $n = 7$).

$$\langle H_B \rangle \approx \frac{1}{2} \cdot 10^{-24} \cdot 2^{36 \cdot 49} = \frac{1}{2} \cdot 10^{-24} \cdot 2^{1764} \approx 10^{507} \text{ J}.$$

Already for a two-digit modulus the energy exceeds the mass-energy of the observable universe ($\sim 4 \times 10^{69}$ J) by a factor of about 10^{437} .

Three-digit number ($N \leq 1000$, $n = 10$).

$$\langle H_B \rangle \approx \frac{1}{2} \cdot 10^{-24} \cdot 2^{36 \cdot 100} = \frac{1}{2} \cdot 10^{-24} \cdot 2^{3600} \approx 10^{1059} \text{ J}.$$

For comparison, the total mass-energy of the observable universe is approximately 4×10^{69} J. The energy required to factor a three-digit number with this protocol exceeds the universe's mass-energy by a factor of roughly $\boxed{10^{990}}$ (that is, by about 990 orders of magnitude).

Eight-digit number ($N \leq 10^8$, $n = 27$).

$$\langle H_B \rangle \approx \frac{1}{2} \cdot 10^{-24} \cdot 2^{36 \cdot 729} = \frac{1}{2} \cdot 10^{-24} \cdot 2^{26244} \approx 10^{7876} \text{ J}.$$

This is an astronomical quantity that does not admit a meaningful physical interpretation.

How steep is the growth? Because the exponent grows as n^2 , one extra bit of N multiplies the energy by $2^{36((n+1)^2 - n^2)} = 2^{36(2n+1)}$ — a factor $\sim 2^{180} \approx 10^{54}$ already in passing from $n = 2$ to $n = 3$. The cost therefore vaults across every physical energy scale within a single bit of N : a one-digit modulus with $n = 2$ ($N \leq 4$) costs $\sim 10^{19}$ J (comparable to the explosive yield of the world's nuclear arsenals), whereas $n = 3$ ($N \leq 8$) already costs $\sim 10^{73}$ J — some three orders of magnitude beyond the mass-energy of the observable universe. There is essentially no integer N whose cost lands at an intermediate “stellar” ($\sim 10^{47}$ J) or “galactic” ($\sim 10^{59}$ J) scale; the n^2 in the exponent steps right over them.

6.5. Energy levels and occupation numbers

The estimates above are mean energies; it is worth restating them in terms of the oscillators' quantized energy levels, which is the physically natural measure of how far the required states lie from anything realizable.

Each mode is a harmonic oscillator with Hamiltonian $H = \hbar\omega(\hat{N} + \frac{1}{2})$ and discrete spectrum

$$E_k = \hbar\omega\left(k + \frac{1}{2}\right), \quad k = 0, 1, 2, \dots,$$

of uniform level spacing $\hbar\omega$; the eigenstates $|k\rangle$ are the Fock (number) states. None of the states the algorithm uses — squeezed vacua, approximate position eigenstates, or GKP combs — is an energy eigenstate; each is a superposition of many Fock states. The energies of §6.3 therefore correspond to a *mean* occupation number

$$\langle \hat{N} \rangle = \frac{\langle H \rangle}{\hbar\omega} - \frac{1}{2}.$$

With the Table I parameters (dropping the negligible $-\frac{1}{2}$),

$$\langle \hat{N}_A \rangle \approx \frac{1}{2} 2^{32n}, \quad \langle \hat{N}_B \rangle \approx \frac{1}{2} 2^{36n^2}, \quad \langle \hat{N}_C \rangle \approx \frac{1}{4} 2^{100n}.$$

The dominant mode B must sit, on average, at occupation number $\langle \hat{N}_B \rangle \approx \frac{1}{2} 2^{36n^2}$: for a three-digit modulus ($n = 10$) this is about $2^{3599} \approx 10^{1083}$, i.e. the oscillator would have to be excited to roughly its 10^{1083} -th level.

The excitation is, moreover, not concentrated at a single level. For a squeezed vacuum of parameter s (so $\Delta = e^{-s}$ and $\langle \hat{N} \rangle = \sinh^2 s$ — mode C is exactly of this form), the photon-number distribution contains only even Fock states and has variance

$$\text{Var}(\hat{N}) = 2 \sinh^2 s \cosh^2 s = \frac{1}{2} \sinh^2(2s) \approx 2 \langle \hat{N} \rangle^2,$$

so the spread $\Delta N \approx \sqrt{2} \langle \hat{N} \rangle$ is of the same order as the mean itself. The GKP states of modes A and B are likewise broad superpositions in Fock space. Each oscillator is therefore not merely raised to an astronomically high level but coherently delocalized across a band of $\sim \langle \hat{N} \rangle$ levels, from the ground state up to occupation numbers of order 2^{36n^2} .

This is the level-occupation counterpart of the energy estimates, and it sharpens the sense in which the protocol is unphysical: state-of-the-art squeezing reaches mean occupation numbers of order ten (about 15 dB), whereas mode B for even a one-digit N already demands occupation numbers far beyond what any real cavity can hold.

6.6. Discussion: a fundamental tradeoff

Why is the energy cost so high? The three oscillators function as continuous registers carrying arbitrarily-precise real numbers. Storing 2^{n^2} distinguishable values in a single oscillator requires squeezing by a factor $2^{-n^2/2}$, which costs energy growing as 2^{n^2} .

This is not a quirk of the protocol; it is a fundamental tradeoff. The Heisenberg uncertainty $\Delta Q \cdot \Delta P \geq 1/2$ means that a register of size S in Q -space, with precision δ , requires $\Delta P \geq 1/(2\delta)$, hence energy at least $\sim 1/\delta^2$. To store the value $a^x \bmod N$ for an integer a^x that can be as large as $N^m \sim 2^{n \cdot 17n} = 2^{17n^2}$, mode B needs distinguishable position values up to $\sim N^m$, with precision ~ 1 , hence energy at least $\sim N^{2m} = 2^{34n^2}$. This matches our calculation up to constants.

By contrast, Shor's algorithm uses $O(\log N) = O(n)$ qubits, each at constant energy $\sim \hbar\omega$, for a total of $O(n\hbar\omega)$ — polynomial in n .

The 3-oscillator-1-qubit algorithm thus trades qubit *count* (which scales as n) for energy per mode (which scales as 2^{n^2}). The total cost is similar in some abstract complexity-theoretic sense (both are polynomial in some natural resource), but the algorithm is not a practical alternative to Shor — it is a striking demonstration that the qubit count can in principle be made constant, at the price of squeezing energy that scales as $2^{\Theta(n^2)}$.

7. Conclusion

We have analyzed the factoring algorithm of [1] in detail. The algorithm uses three continuous-variable oscillators and a single qubit to implement, in polynomial time, the same period-finding procedure as Shor’s algorithm. The key technical innovations are:

- **Coherent repeated squaring** on a single oscillator using the V_α gate, which peels one bit of A and conditionally multiplies B by a precomputed power of a at each step.
- **The lattice-state trick** on mode B : preparing B in $M_N |\text{GKP}\rangle$ enforces modular structure geometrically; the real-arithmetic gates then automatically compute modulo N .
- **The conjugation trick** converting multiplication by $f_{a,N,m}(x)$ on B into a controlled translation, the CV analogue of Shor’s modular-exponentiation unitary.
- **Approximate function evaluation** bounds: the only source of approximation error in the discrete-on-continuous arithmetic is the LSB gate’s sensitivity to peak-width deviation from integers; this error can be made exponentially small.

On the practical side, the energy budget of $2^{\Theta(n^2)}$ rules out the algorithm as a candidate quantum factoring scheme. The fundamental tradeoff between qubit count and energy per oscillator is intrinsic and cannot be circumvented by clever circuit design: it follows from the Heisenberg uncertainty principle and the size of the integers being computed.

The paper’s contribution should therefore be read as a complexity-theoretic statement about *which* resources can be substituted for which: continuous-variable squeezing can in principle replace qubit count, but only at exponentially higher energy cost.

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Appendix A. Auxiliary derivations

A.1. Wavefunction of the vacuum

In the position representation, $a = (Q + iP)/\sqrt{2} = (x + d/dx)/\sqrt{2}$. The condition $a\Psi_{\text{vac}} = 0$ becomes $d\Psi_{\text{vac}}/dx = -x\Psi_{\text{vac}}$, with solution $\Psi_{\text{vac}}(x) = Ce^{-x^2/2}$. Normalizing via $\int |\Psi_{\text{vac}}|^2 dx = 1$ gives $C = \pi^{-1/4}$.

A.2. Wavefunction form of M_α

For any $\Psi \in L^2(\mathbb{R})$:

$$(M_\alpha \Psi)(x) = \langle x | M_\alpha \Psi \rangle = \int dx' \langle x | M_\alpha | x' \rangle \Psi(x') = \int dx' \langle x | \alpha x' \rangle \Psi(x') = \int dx' \delta(x - \alpha x') \Psi(x').$$

Substituting $u = \alpha x'$ ($dx' = du/\alpha$):

$$(M_\alpha \Psi)(x) = \int \frac{du}{\alpha} \delta(x - u) \Psi(u/\alpha) = \frac{1}{\alpha} \Psi(x/\alpha).$$

The factor of $\frac{1}{\sqrt{\alpha}}$ rather than $\frac{1}{\alpha}$ comes from working with normalized improper states (sometimes encoded in the convention $M_\alpha |x\rangle = \sqrt{\alpha} |\alpha x\rangle$); both conventions are equivalent up to normalization factors absorbed elsewhere.

A.3. Wavefunction form of e^{iP}

Since $P = -i d/dx$, we have $e^{iP} = e^{d/dx}$. For analytic Ψ , the Taylor expansion of $\Psi(x + h)$ around $h = 0$ is

$$\Psi(x + h) = \sum_{n=0}^{\infty} \frac{h^n}{n!} \frac{d^n \Psi}{dx^n}(x).$$

At $h = 1$: $\Psi(x + 1) = e^{d/dx} \Psi(x) = (e^{iP} \Psi)(x)$. For position eigenstates: $(e^{iP} \delta_{x'})(x) = \delta(x + 1 - x') = \delta(x - (x' - 1))$, hence $e^{iP} |x'\rangle = |x' - 1\rangle$.

A.4. Rotation identity for the squeezing operator

We show $RS(z)R^\dagger = S(-z)$ for $R = e^{-i\pi\hat{N}/2}$ where $\hat{N} = (Q^2 + P^2 - 1)/2$. The phase-space rotation generated by $\hat{N}$ satisfies

$$RQR^\dagger = P, \quad RPR^\dagger = -Q$$

(a quarter-turn in phase space). The generator of $S(z)$ is $G = (QP + PQ)/2$; under conjugation:

$$RG R^\dagger = \frac{1}{2}(RQR^\dagger \cdot RPR^\dagger + RPR^\dagger \cdot RQR^\dagger) = \frac{1}{2}(P(-Q) + (-Q)P) = -G.$$

Hence $Re^{izG}R^\dagger = e^{-izG}$, i.e. $RS(z)R^\dagger = S(-z)$.

A.5. Scaling action of M_N on Gaussian peaks

The wavefunction of $|\chi_\Delta(z)\rangle$ is $\Psi_\Delta(x - z) = (\pi\Delta^2)^{-1/4} e^{-(x-z)^2/(2\Delta^2)}$. Apply $(M_N \Psi)(x) = \frac{1}{\sqrt{N}} \Psi(x/N)$:

$$\begin{aligned} (M_N \chi_\Delta(z))(x) &= \frac{1}{\sqrt{N}(\pi\Delta^2)^{1/4}} e^{-(x/N - z)^2/(2\Delta^2)} \\ &= \frac{1}{(\pi(N\Delta)^2)^{1/4}} e^{-(x - Nz)^2/(2(N\Delta)^2)} = \chi_{N\Delta}(Nz)(x), \end{aligned}$$

using $(x/N - z)^2/\Delta^2 = (x - Nz)^2/(N\Delta)^2$ and combining the prefactors.

A.6. Truncated peak orthogonality

The wavefunctions are $\chi_\Delta^\varepsilon(z)(x) = \mathcal{N}^{-1} \mathbf{1}_{[z-\varepsilon, z+\varepsilon]}(x) \Psi_\Delta(x-z)$. The inner product is

$$\langle \chi_\Delta^\varepsilon(z) | \chi_\Delta^\varepsilon(z') \rangle = \mathcal{N}^{-2} \int_{\mathbb{R}} \mathbf{1}_{[z-\varepsilon, z+\varepsilon]}(x) \mathbf{1}_{[z'-\varepsilon, z'+\varepsilon]}(x) \overline{\Psi_\Delta(x-z)} \Psi_\Delta(x-z') dx.$$

For $z, z' \in \mathbb{Z}$ with $z \neq z'$, the indicator-function product vanishes everywhere (the intervals are disjoint since $|z - z'| \geq 1 > 2\varepsilon$). For $z = z'$, normalization gives 1.

A.7. Inductive proof for $V_{a,N,m}$

Claim. After j applications of $V_{a^{2^{j-1}} \bmod N}$, the state is

$$|\Psi_j\rangle = |x^{(j)}\rangle |y^{(j)}\rangle |z^{(j)}\rangle |0\rangle,$$

with $x^{(j)} = (x - \sum_{i < j} x_i 2^i) / 2^j$, $y^{(j)} = \prod_{i < j} (a^{2^i} \bmod N)^{x_i} \cdot y$, $z^{(j)} = 2^j z + \sum_{i < j} 2^{j-1-i} x_i$.

Base case. $j = 0$: $|\Psi_0\rangle = |x\rangle |y\rangle |z\rangle |0\rangle$. ✓

Inductive step. Apply V_α with $\alpha = a^{2^j} \bmod N$. The LSB of $x^{(j)} = \sum_{k \geq j} x_k 2^{k-j}$ is x_j . By Theorem 3.4:

- $x^{(j+1)} = (x^{(j)} - x_j) / 2 = (x - \sum_{i \leq j} x_i 2^i) / 2^{j+1}$, ✓
- $y^{(j+1)} = (a^{2^j} \bmod N)^{x_j} \cdot y^{(j)} = \prod_{i \leq j} (a^{2^i} \bmod N)^{x_i} \cdot y$, ✓
- $z^{(j+1)} = 2z^{(j)} + x_j = 2^{j+1}z + \sum_{i \leq j} 2^{j-i} x_i$. ✓

After m steps, B holds $f_{a,N,m}(x) \cdot y$. The cleanup pass $(V_1^\dagger)^m$ reverses the bit-transfer step by step (Corollary 3.5), restoring A and C to their input values.

Appendix B. Error analysis for the physical algorithm

B.1. Single-LSB-gate error

Suppose Ψ has support within ε of integer n . Write $y = n + \delta$ with $|\delta| \leq \varepsilon$. After the LSB circuit:

- After H_Q : $\Psi(y) \otimes (|0\rangle + (-1)^b |1\rangle) / \sqrt{2}$.
- After $\text{ctrl-}e^{i\pi Q_A}$: using $e^{i\pi y} = (-1)^{n_0} \cdot e^{i\pi \delta}$ (with $n_0 = n \bmod 2$):

$$\Psi(y) \otimes \frac{|0\rangle + (-1)^{b+n_0} e^{i\pi \delta} |1\rangle}{\sqrt{2}}.$$

- After H_Q (setting $c = b \oplus n_0$): $\Psi(y) \otimes \left[\frac{1+(-1)^c e^{i\pi \delta}}{2} |0\rangle + \frac{1-(-1)^c e^{i\pi \delta}}{2} |1\rangle \right]$.

The ideal state is $\Psi(y) \otimes |c\rangle$. The error norm squared is

$$\int |\Psi(y)|^2 \cdot \frac{|1 - e^{i\pi \delta}|^2}{2} dy \leq \int |\Psi(y)|^2 \cdot \frac{(\pi \delta)^2}{2} dy \leq \frac{\pi^2 \varepsilon^2}{2}.$$

Thus $\|U^{\text{LSB}}(\Psi \otimes |b\rangle) - \Psi \otimes |b \oplus n_0\rangle\| \leq \pi \varepsilon / \sqrt{2}$.

B.2. Error in V_α and $V_{a,N,m}$

The single-gate bound of §B.1 ($\pi\varepsilon/\sqrt{2}$) controls the deviation of one localized packet. The trace-distance error of the *composite* gate, however, is governed by the “approximately computes” framework of [1]: a per-point overlap deficit ε yields a global function-evaluation error $6\varepsilon^{1/4}$ (Lemma 5.3 of [1], built on the overlap bound $|\langle\Phi, \Phi'\rangle| \geq 1 - 2\varepsilon$ of Lemma 5.2), the fourth root arising from the cross-terms between correct and incorrect qubit outcomes together with the overlap-to-norm conversion.

Each V_α applies two such LSB gates — on mode A (support width ε_A) and on mode C (support width $2\varepsilon_C$ after M_2) — giving $\delta_{V_\alpha}(\varepsilon_A, \varepsilon_C) = 7\varepsilon_A^{1/4} + 7(2\varepsilon_C)^{1/4}$ (Lemma 5.5 of [1]). Composing the m -step chain and the cleanup pass $(V_1^\dagger)^m$ (using that the A-width halves and the C-width doubles at each step) gives $\delta_{V_{a,N,m}}(\varepsilon_A, \varepsilon_C) = 88\varepsilon_A^{1/4} + 88(2^m\varepsilon_C)^{1/4}$ (Lemma 5.7), and the conjugation $U_{a,N,m} = V_{a,N,m} \circ e^{-iP_B} \circ V_{a,N,m}^\dagger$ doubles it (Lemma 5.10):

$$\delta_{U_{a,N,m}}(\varepsilon_A, \varepsilon_C) = 176\varepsilon_A^{1/4} + 176(2^m\varepsilon_C)^{1/4}.$$

Substituting $\varepsilon_A = \sqrt{\Delta_A} = 2^{-8n}$ and $\varepsilon_C = \sqrt{\Delta_C} = 2^{-25n}$ from Table I (so $2^m\varepsilon_C = 2^{17n} \cdot 2^{-25n} = 2^{-8n}$, with $m = 17n$), both fourth roots equal 2^{-2n} , so

$$\delta_{U_{a,N,m}} = 176 \cdot 2^{-2n} + 176 \cdot 2^{-2n} = 352 \cdot 2^{-2n},$$

which is Lemma 5.11 of [1], the dominant contribution to Proposition 5.5.

B.3. Approximate translation-invariance

For $|\Psi\rangle = M_N |\text{GKP}_{\kappa_B, \Delta_B}\rangle \propto \sum_{y \in N\mathbb{Z}} \eta_{\kappa_B/N}(y) |\chi_{N\Delta_B}(y)\rangle$ and translation $e^{-i\ell NP_B}$:

$$e^{-i\ell NP_B} |\Psi\rangle \propto \sum_{y' \in N\mathbb{Z}} \eta_{\kappa_B/N}(y' - \ell N) |\chi_{N\Delta_B}(y')\rangle.$$

By peak orthogonality (Lemma 2.12, after truncation):

$$\langle \Psi | e^{-i\ell NP_B} | \Psi \rangle = \frac{\sum_{y \in N\mathbb{Z}} \eta_{\kappa_B/N}(y) \eta_{\kappa_B/N}(y - \ell N)}{\sum_{y \in N\mathbb{Z}} |\eta_{\kappa_B/N}(y)|^2}.$$

Completing the square: $y^2 + (y - \ell N)^2 = 2(y - \ell N/2)^2 + \ell^2 N^2/2$, giving

$$\eta_{\kappa_B/N}(y) \eta_{\kappa_B/N}(y - \ell N) = \frac{\kappa_B/N}{\sqrt{\pi}} e^{-(\kappa_B/N)^2(y - \ell N/2)^2} \cdot e^{-(\kappa_B \ell)^2/4}.$$

Approximating the sum by an integral divided by the spacing N :

$$\sum_{y \in N\mathbb{Z}} e^{-(\kappa_B/N)^2(y - \ell N/2)^2} \approx \frac{1}{N} \cdot \frac{\sqrt{\pi}}{\kappa_B/N} = \frac{\sqrt{\pi}}{\kappa_B}.$$

Numerator $\approx \frac{1}{N} e^{-(\kappa_B \ell)^2/4}$; denominator $\approx \frac{1}{N}$. Hence $\langle \Psi | e^{-i\ell NP_B} | \Psi \rangle \approx e^{-(\kappa_B \ell)^2/4}$.

The corresponding state difference satisfies $\|e^{-i\ell NP_B} |\Psi\rangle - |\Psi\rangle\|^2 \leq (\kappa_B \ell)^2/2$. With $|\ell| \leq N^{m-1} \leq 2^{n(17n-1)}$ and $\kappa_B = 2^{-18n^2}$:

$$|\kappa_B \ell| \leq 2^{-18n^2 + n(17n-1)} = 2^{-n^2 - n}.$$

The error is $2^{-\Theta(n^2)}$.

B.4. Combining errors: Proposition 5.5

Following [1] (Table II), one introduces intermediate states $\Psi_a = \Psi^{(0)}, \Psi^{(1)}, \dots, \Psi^{(5)} = \Phi_a$ and bounds the consecutive trace distances $\|\Psi^{(j)} - \Psi^{(j-1)}\|_1 \leq \varepsilon^{(j)}$:

Transition	Source of error	$\varepsilon^{(j)}$
$\Psi^{(0)} \rightarrow \Psi^{(1)}$	truncate A (peaks + window), B (peaks), C	$9 \cdot 2^{-2n}$ (Lemma 4.1 of [1])
$\Psi^{(1)} \rightarrow \Psi^{(2)}$	approximate function evaluation of $U_{a,N,m}$	$352 \cdot 2^{-2n}$ (Lemma 5.11 of [1])
$\Psi^{(2)} \rightarrow \Psi^{(3)}$	replace $f_{a,N,m}(z)$ -shift by $(a^z \bmod N)$ -shift	2^{-2n} (Lemma 4.2 of [1])
$\Psi^{(3)} \rightarrow \Psi^{(4)}$	restore mode-A envelope to all of $\mathbb{Z}$	2^{-2n} (Lemma 4.3 of [1])
$\Psi^{(4)} \rightarrow \Psi^{(5)}$	restore mode-A peaks ($\chi^{\varepsilon_A} \rightarrow \chi$)	2^{-2n} (Lemma 4.4 of [1])

By the triangle inequality,

$$\|\Psi_a - \Phi_a\|_1 \leq \sum_{j=1}^5 \varepsilon^{(j)} = (9 + 352 + 1 + 1 + 1) \cdot 2^{-2n} = 364 \cdot 2^{-2n}.$$

The dominant contribution is the function-evaluation term $352 \cdot 2^{-2n}$ (§B.2); every term scales as 2^{-2n} , and the prefactor 364 is the sum of the five constants.

Appendix C. Fourier analysis details

C.1. Poisson summation

For $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ with Fourier transform $\hat{f}(w) = \int f(x) e^{-2\pi i w x} dx$:

$$\sum_{n \in \mathbb{Z}} f(x + n) = \sum_{k \in \mathbb{Z}} \hat{f}(k) e^{2\pi i k x}.$$

Applied to a Dirac comb of spacing r : $\sum_{\ell \in \mathbb{Z}} \delta(x - \ell r)$ has Fourier transform $\frac{1}{r} \sum_{n \in \mathbb{Z}} \delta(w - n/r)$. The shifted comb $\sum_{\ell} \delta(x - j - \ell r)$ has Fourier transform $e^{-2\pi i j w} \cdot \frac{1}{r} \sum_n \delta(w - n/r)$.

C.2. Suitability of p_{Φ_a} (Proposition 5.7)

The distribution $p_{\Phi_a}(w) \approx |\widehat{\Phi_a^{(0)}}(w)|^2$ is, up to a Gaussian smearing of width κ_A , a uniform distribution on $\frac{1}{r}\mathbb{Z}$. Around each lattice point d/r , the Gaussian peak has width $\kappa_A = 2^{-16n}$ and area $\Omega(1/r)$.

The set $\Gamma_d(a)$ consists of windows of width $1/q < 1/N^2 = 2^{-2n}$ around $d/r + \mathbb{Z}$. Since $\kappa_A = 2^{-16n} \ll 1/q = 2^{-2n}$ (the peak is narrower than the window), essentially all of each Gaussian's mass lies inside its window:

$$\int_{\Gamma_d(a)} p_{\Phi_a}(w) dw \geq (1 - \operatorname{erfc}(1/(2q\kappa_A))) \cdot \Omega(1/r) = \Omega(1/r).$$

C.3. Continued fractions and period extraction

Given $w \in [0, 1)$, the continued fraction algorithm computes:

1. Set $\alpha_0 = w$.

2. For $i = 0, 1, 2, \dots$: set $a_i = \lfloor \alpha_i \rfloor$ and $\alpha_{i+1} = 1/(\alpha_i - a_i)$ (terminating if $\alpha_i = a_i$).
3. Convergents p_i/q_i are defined by the recurrence $p_i = a_i p_{i-1} + p_{i-2}$, $q_i = a_i q_{i-1} + q_{i-2}$ with $p_{-1} = 1$, $p_{-2} = 0$, $q_{-1} = 0$, $q_{-2} = 1$ (so that $p_0 = a_0$ and $q_0 = 1$; here $a_0 = 0$ since $w \in [0, 1)$).

For each convergent p_i/q_i in lowest terms, $|w - p_i/q_i| < 1/(q_i q_{i+1}) \leq 1/q_i^2$. If $|w - n/r| < 1/(2r^2)$ for some n, r with $\gcd(n, r) = 1$, then n/r appears as a convergent.

Testing each convergent: reject if $q_i \geq N$ or if $a^{q_i} \not\equiv 1 \pmod{N}$; accept the smallest q_i that satisfies $a^{q_i} \equiv 1 \pmod{N}$. The first convergent satisfying both conditions equals r iff $\gcd(n, r) = 1$.

Probability of success per shot. Conditional on w being uniformly drawn from $\{0/r, 1/r, \dots, (r-1)/r\}$, the probability that the corresponding n is coprime to r is $\phi(r)/r$, where ϕ is Euler's totient. By a classical estimate, $\phi(r)/r \geq \Omega(1/\log \log r)$, so $O(\log \log N)$ repetitions suffice with constant failure probability.

Appendix D. Energy of an approximate GKP state

We derive the formula

$$\langle H \rangle_{\text{GKP}_{\kappa, \Delta}} \approx \frac{\hbar\omega}{4} \left(\frac{1}{\Delta^2} + \frac{1}{\kappa^2} \right)$$

used in Section 6, valid in the regime of tight squeezing $\Delta, \kappa \ll 1$.

D.1. Setup

The Hamiltonian of a single bosonic mode is

$$H = \hbar\omega \left(\hat{N} + \frac{1}{2} \right) = \frac{\hbar\omega}{2} (Q^2 + P^2),$$

where we have used $\hat{N} = \frac{1}{2}(Q^2 + P^2 - 1)$; the identity is exact, with the $\frac{\hbar\omega}{2}$ zero-point energy already included in $\frac{\hbar\omega}{2}(Q^2 + P^2)$. The mean energy of any state $|\Psi\rangle$ is thus

$$\langle H \rangle = \frac{\hbar\omega}{2} (\langle Q^2 \rangle + \langle P^2 \rangle).$$

The approximate GKP state $|\text{GKP}_{\kappa, \Delta}\rangle$ is, up to normalization, a Dirac comb at integer positions, where each peak has been smeared into a Gaussian of width Δ in position, and the entire comb has been multiplied by a Gaussian envelope of width $1/\kappa$ in position. Equivalently, it has Gaussian structure in *both* quadratures: by Poisson summation the position-space comb (peaks of width Δ under an envelope of width $1/\kappa$) is, in momentum space, a comb with peaks of width κ under an envelope of width $1/\Delta$ [3]. We use this dual description to read off the second moments.

For ease of calculation we work with the equivalent description: the variances $\langle Q^2 \rangle$ and $\langle P^2 \rangle$ of the approximate GKP state decompose into independent contributions from the peak and envelope structures, which we now compute separately.

D.2. Heuristic decomposition via two independent Gaussians

A clean way to derive the result is to recognize that the position-space wavefunction of an approximate GKP state has the form

$$\chi(x) \propto \sum_{z \in \mathbb{Z}} \eta_{\kappa}(z) \Psi_{\Delta}(x - z),$$

where $\Psi_\Delta(x-z)$ is a Gaussian peak of width Δ centered at z , and $\eta_\kappa(z)$ is a Gaussian envelope of width $1/\kappa$. For $\kappa \ll 1$ (envelope much wider than the inter-peak spacing of 1), the discrete sum can be approximated by a continuous convolution, and the state's position probability density factorizes (approximately) as a convolution of two Gaussians:

- the *envelope* Gaussian of variance $\sigma_{\text{env}}^2 = 1/(2\kappa^2)$,
- the *peak* Gaussian of variance $\sigma_{\text{peak}}^2 = \Delta^2/2$.

By the standard rule for convolved Gaussians, the position variance is the sum:

$$\langle Q^2 \rangle = \sigma_{\text{env}}^2 + \sigma_{\text{peak}}^2 = \frac{1}{2\kappa^2} + \frac{\Delta^2}{2}.$$

The momentum-space wavefunction is the Fourier transform of χ . By the standard duality of Gaussians under Fourier transform — a Gaussian of width σ in position becomes a Gaussian of width $1/(2\sigma)$ in (angular) momentum — the roles of κ and Δ are swapped:

- peak Gaussians of position width Δ have momentum width $1/\Delta$, giving momentum variance $1/(2\Delta^2)$;
- envelope of position width $1/\kappa$ has momentum width κ , giving momentum variance $\kappa^2/2$.

Thus:

$$\langle P^2 \rangle = \frac{1}{2\Delta^2} + \frac{\kappa^2}{2}.$$

D.3. Combining

Substituting into $\langle H \rangle = \frac{\hbar\omega}{2}(\langle Q^2 \rangle + \langle P^2 \rangle)$:

$$\langle H \rangle = \frac{\hbar\omega}{2} \left(\frac{1}{2\Delta^2} + \frac{\Delta^2}{2} + \frac{1}{2\kappa^2} + \frac{\kappa^2}{2} \right) = \frac{\hbar\omega}{4} \left(\frac{1}{\Delta^2} + \frac{1}{\kappa^2} + \Delta^2 + \kappa^2 \right).$$

For small Δ and κ (tight squeezing in both quadratures), the inverse-square terms dominate the linear-square terms, giving the asymptotic result:

$$\langle H \rangle_{\text{GKP}_{\kappa,\Delta}} \approx \frac{\hbar\omega}{4} \left(\frac{1}{\Delta^2} + \frac{1}{\kappa^2} \right) = \text{Energy of an approximate GKP state (tight-squeezing limit)}.$$

D.4. Interpretation

The two terms have a clean physical meaning via the Heisenberg uncertainty principle:

- The term $\frac{1}{\Delta^2}$ comes from the requirement that each peak be tightly localized in position (width Δ). By uncertainty, this forces a large momentum spread $\sim 1/\Delta$, hence a large kinetic energy $\sim P^2 \sim 1/\Delta^2$.
- The term $\frac{1}{\kappa^2}$ comes from the requirement that the comb be spatially broad (envelope width $1/\kappa$). Each peak in the comb has positions of magnitude up to $\sim 1/\kappa$, so the potential energy $\sim Q^2 \sim 1/\kappa^2$ is correspondingly large.

The total energy is therefore essentially $\hbar\omega/4$ times the sum of the squared spreads in both quadratures, exactly as one would expect for any Gaussian state. The GKP state is special only in that both spreads must be tightly controlled simultaneously, leading to the additive contribution of two large quantities.

Appendix E. Circuit visualisation of the gates

Notation & functionality	Implementation
(a) $ x\rangle \xrightarrow{M_\alpha} \alpha \cdot x\rangle$	$\underbrace{S(-(\log \alpha)/\ell) \cdots S(-(\log \alpha)/\ell)}_{\times \ell \text{ where } \ell := \lceil \log \alpha \rceil}$
(b) $ x\rangle \xrightarrow{e^{-iRP}} x + R\rangle$	$M_{1/R} \xrightarrow{e^{-iP}} M_R$
(c) $ b\rangle \xrightarrow{\text{ctrl-}M_\alpha} \alpha^b \cdot x\rangle$	$M_{\sqrt{\alpha}} \xrightarrow{e^{-i\pi \hat{N}/2}} M_{\sqrt{\alpha}}^\dagger \xrightarrow{e^{i\pi \hat{N}/2}}$
(d) $ x\rangle \xrightarrow{\text{LSB}} x\rangle$ $ b\rangle \xrightarrow{\oplus} x_0 \oplus b\rangle$	$e^{i\pi Q} \xrightarrow{H} H$
(e) V_α acting on $ x\rangle, y\rangle, z\rangle, 0\rangle$ to produce $ (x - x_0)/2\rangle, \alpha^{x_0} \cdot y\rangle, 2z + x_0\rangle, 0\rangle$	$M_{1/2} \xrightarrow{\text{LSB}} M_\alpha \xrightarrow{M_2} \text{LSB}$
(f) $V_{a,N,m}$ acting on $ x\rangle, y\rangle, z\rangle, 0\rangle$ to produce $ x\rangle, y \cdot f_{a,N,m}(x)\rangle, z\rangle, 0\rangle$	$V_{a^{2^0 \bmod N}} V_{a^{2^1 \bmod N}} \cdots V_{a^{2^{m-1} \bmod N}} (V_1^\dagger)^m$
(g) $U_{a,N,m}$ acting on $ x\rangle, y\rangle, z\rangle, 0\rangle$ to produce $ x\rangle, y + f_{a,N,m}(x)\rangle, z\rangle, 0\rangle$	$V_{a,N,m}^\dagger \xrightarrow{e^{-iP}} V_{a,N,m}$

Figure 5: The gates of the algorithm. Left: notation and action on basis states. Right: circuit implementation. From top to bottom, the primitive gates M_α , e^{-iRP} , ctrl- M_α and U^{LSB} , followed by the composite gates V_α , $V_{a,N,m}$ and $U_{a,N,m}$. Reproduced from [?].